Tourism Pressure and Municipal Electricity Demand: Mean Effects, Extreme Dependence and Vulnerability Assessment

Pietro Colombo Consuelo Maria Nava Camilla Andreozzi Jethro Browell

May 2026

Abstract

This paper studies how tourism is transmitted into municipal electricity demand in Valle d'Aosta using a high-resolution municipality-month panel. The empirical analysis is organized in three linked blocks. First, we estimate nested mixed-effects models to isolate the conditional mean effect of tourism arrivals on electricity consumption after controlling for municipality heterogeneity, meteorology, seasonality and persistence. Second, we study residual tail dependence with copulas and municipality-level co-exceedance measures to identify where tourism and electricity shocks jointly concentrate in the upper tail. Third, we combine the historical tail-risk diagnostics with q95 tourism scenarios to produce forward stress rankings and a validation exercise based on realized 2025 positive stress. The results show a robust positive association between tourism and the conditional mean of electricity demand, stronger transmission in tourism-intensive municipalities, weak pooled tail dependence, and localized extreme vulnerabilities concentrated in a limited subset of municipalities. The analysis is observational, so these estimates are conditional-mean associations rather than causal effects. The scenario exercise adds that these vulnerabilities are informative for prioritization, but not for exact point prediction of stress magnitude.

Keywords: tourism demand; electricity consumption; mixed-effects models; copulas; tail dependence; municipal panel data

1 Introduction

Valle d'Aosta, an autonomous region in north-western Italy, combines a small and sparsely populated territory with a highly tourism-oriented local economy. Tourist arrivals reached about 1.585 million in 2025, underscoring the intensity of visitor pressure relative to the size of the region. Tourism is widely recognized as a key driver of regional activity, especially through winter sports, hospitality, transport, and other visitor-related services. This makes the region a useful setting for studying how tourism pressure translates into electricity demand and whether that relationship is concentrated in particular municipalities or periods of stress. The case is substantively important because Valle d'Aosta is characterized by strong seasonality, pronounced spatial heterogeneity, and a mountainous geography that can amplify local demand fluctuations. In this setting, electricity demand is shaped not only by resident population and weather, but also by highly mobile and spatially concentrated tourist flows. Understanding these interactions is therefore relevant for energy planning, infrastructure adequacy, and local resilience in the face of growing tourism pressure and climate-related uncertainty.

The broader national energy context reinforces this motivation. According to Terna, Italy's electricity demand in 2024 was 311.9 TWh, of which 260.9 TWh were covered by domestic production and 51.0 TWh by net imports, implying that 16.4% of electricity demand was met from abroad [Terna, 2025]. At the same time, the IEA reports that Italy's total energy supply in 2024 remained dominated by natural gas (39.8%) and oil products (35.6%), with smaller contributions from biofuels and waste (11.2%), solar, wind and other renewables (8.1%), hydropower (3.6%), and coal (1.8%) [International Energy Agency, 2025]. This combination of external dependence and fossil-fuel exposure makes Italy particularly sensitive to international energy shocks and price volatility, strengthening the policy relevance of understanding localized electricity stress in tourism-intensive regions.

This paper investigates how tourism affects electricity demand, what are the main drivers of that relationship, and where tourism-related electricity stress is most likely to occur. Moreover, it provides a framework for forecasting and it design a scenario-based analysis of how tourism demand translates into electricity stress.

From a policy perspective, this study matters for at least three reasons. First, electricity systems in tourism-oriented mountain regions must cope with large seasonal demand swings that are not driven only by resident population or weather, but also by highly mobile and spatially concentrated visitor flows. Second, if tourism demand translates into electricity stress unevenly across municipalities, regional planning based only on aggregate demand can understate local bottlenecks in distribution networks, peak-load management, and service reliability. Third, understanding whether tourism affects only expected demand or also the joint upper tail of electricity shocks is directly relevant for resilience policy: infrastructure reinforcement, storage deployment, demand-side flexibility, and emergency planning should be prioritized differently depending on whether the main issue is persistent average pressure or localized extreme co-movement.

This positioning is substantially different from all previous literature on the topic. One of the closest contribution date back to the early 2011s Bakhat and Rossello [2011], who analyse tourism-induced electricity consumption, in the Balearic Islands using aggregate daily load data. However, theier analysis is based on time-series models that disregard a more capillary spatial structure and therefore policy interventions at the municipal level. Most studies on the tourism-energy nexus are either cross-country analyses of aggregate energy use or sectoral decompositions that do not model local electricity demand directly [Badal et al., 2023, Soylu et al., 2024, Bianco, 2020]. Our paper moves beyond that benchmark along three margins that are central for an applied statistical audience. First, we work at the municipality-month level within a single region, which allows the tourism effect to vary over a much finer spatial scale than regional or national studies. Second, we estimate the conditional mean after controlling jointly for demographic composition, meteorology, seasonality, persistence and municipality-specific heterogeneity, rather than inferring electricity implications from broader energy aggregates. Third, we extend the analysis from mean demand pressure to localized extreme dependence, asking not only whether tourism raises electricity demand on average, but also where tourism-related shocks are most likely to coincide with residual electricity stress. Finally, we provide a scenario based forecasting block to translate the mean-effect and tail-risk evidence into a forward-looking policy tool.

The manuscript is organized as follows. Section 2 describes the study setting and data, Section 3 presents the empirical strategy, Section 4 reports the main results, Section 5 discusses the findings.

Taken together, the structure of the paper is deliberately modular. The first empirical block identifies the average tourism effect on electricity demand; the second block isolates residual upper-tail dependence and municipality-level exposure; the third block uses tourism q95 scenarios to translate the historical evidence into forward-looking stress rankings; and the final validation step checks whether the resulting predicted rankings recover realized positive stress in 2025. This separation keeps the interpretation clear: tourism acts both as a mean-demand driver and as a localized amplifier of extreme stress, but these two roles are not identical and are therefore analysed separately.

To make this roadmap concrete, it is useful to anticipate the indicators that the analysis delivers, since each block is summarized by a specific and interpretable quantity. The mean-effect block produces a tourism semi-elasticity, i.e. the percentage change in electricity demand associated with a one-standard-deviation increase in arrivals, together with its heterogeneous counterpart for tourism-intensive municipalities. The tail-risk block produces a region-wide upper-tail dependence coefficient λ_U from the selected copula and, at municipality level, a conditional co-exceedance probability $p_i(0.95)$ and a composite risk score R_i that ranks where extreme tourism and electricity shocks tend to coincide. The forward block, developed in Section 3.1 onwards and central to the section "From Historical Tail Risk to Forward Extreme Stress", then combines these objects into three forward-looking indicators: the scenario load shock ΔL_{it} (the additional demand implied by a q95 tourism scenario), the tail-conditioned impact $I_{it}^{\text{tail}} = \Delta L_{it}\, p_i(0.95)$ (which rescales that shock by historical tail dependence), and the risk-weighted impact $I_{it}^{\text{w}} = \Delta L_{it}\, p_i(0.95)\, R_i$ (which further weights municipalities by structural vulnerability). Their temporal and spatial aggregates yield the regional and municipality-level stress rankings that constitute the paper's main forward-looking output, and whose predictive value is then assessed against realized 2025 stress. Reading the paper through these indicators makes explicit how exposure, dependence and vulnerability are kept analytically distinct and then recombined into a single prioritization tool.

2 Study Setting and Data

The starting dataset is a municipality-month panel for Valle d'Aosta containing 9,546 observations, 71 variables, and 74 municipalities over the period January 2015 to September 2025. The integrated dataset combines electricity demand, tourism arrivals and presences, municipal demographics, and meteorological covariates. The final common modelling sample used in the nested models contains 5,892 observations and 73 municipalities after lag construction and complete-case filtering. The dependent variable is the logarithm of monthly municipal electricity consumption. The main tourism predictor is the logarithm of tourist arrivals. Weather controls include temperature, precipitation, pressure and relative humidity. Calendar controls are represented through Fourier seasonality terms and year effects. Dynamic controls include lagged log consumption at one and twelve months.

The design is observational: tourism arrivals are not randomly assigned across municipalities and months, so the panel does not support a causal identification strategy. Throughout, the estimated tourism coefficients are interpreted as conditional-mean associations after adjustment for municipality heterogeneity, meteorology, seasonality and lagged demand, rather than as causal effects of tourism on electricity consumption.

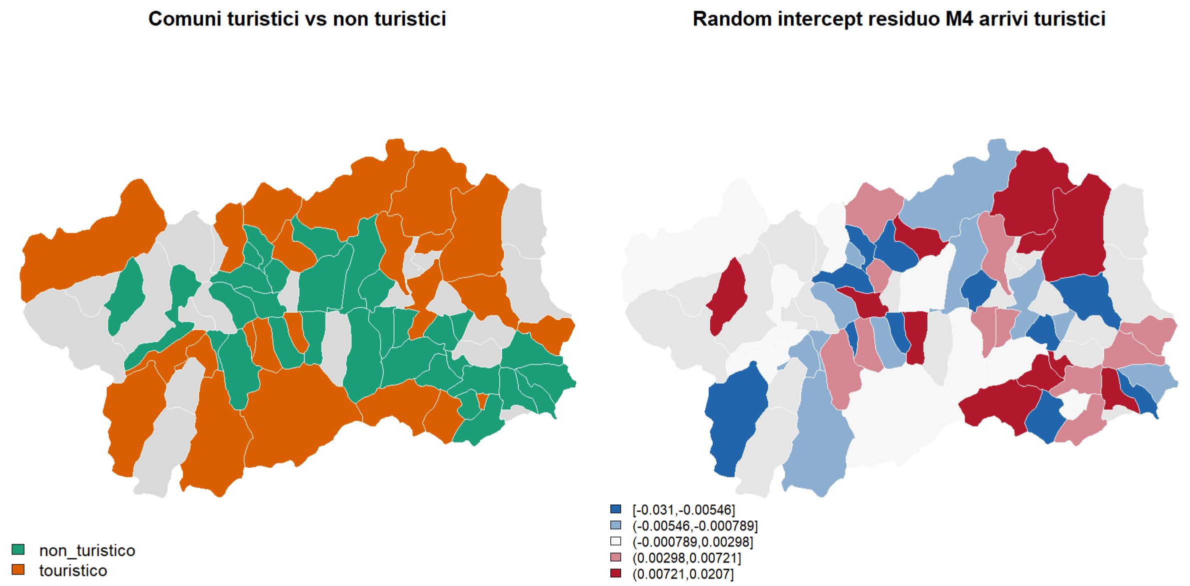

Figure 1: Spatial heterogeneity across municipalities. The report asset summarizes tourism clusters and municipality-level residual heterogeneity from the mean-effect model.

3 Empirical Strategy

The empirical strategy is organized as a sequence of connected steps. We first estimate the conditional mean of municipal electricity demand using progressively richer mixed-effects models so that the tourism coefficient is interpreted after controlling for structural characteristics, meteorology, calendar effects and persistence. We then extract residual variation from the preferred specification and treat the remaining shocks in tourism and electricity demand as the relevant objects for dependence analysis. Finally, we evaluate whether these residual shocks display upper-tail co-movement through parametric copulas and municipality-level co-exceedance measures, allowing us to distinguish broad average tourism pressure from localized extreme vulnerability. This block is a historical analysis of joint tail risk in-sample. We then build forecasting models for tourism demand and use them to generate scenario-based forecasts of electricity demand, with particular attention to municipalities identified as high-risk in the tail analysis. Combining the mean-effect and tail-risk blocks, the strategy delivers a forward-looking tool for scenario analysis and planning.

3.1 Mean-effect model

To understand the average electricity consumption trend in Valle D'Aosta and what are the main drivers of it, we estimate a sequence of nested mixed-effects models [Gelman and Hill, 2007]. The baseline specification includes a municipality-specific random intercept to capture unobserved heterogeneity and a set of fixed effects to control for structural, meteorological, calendar and dynamic predictors. The tourism effect is then added on top of this baseline in two alternative ways: as a homogeneous effect across municipalities, and as an interaction with a tourism intensity indicator to capture heterogeneous transmission. Let y_{it} denote logarithmic electricity consumption in municipality i at month t. The preferred mean-effect specification is the most saturated mixed model in the sequence, namely $M4$:

$$
\begin{aligned}
y_{it} &= \alpha + b_i + \beta_1 \mathrm{ResMean}_i + \beta_2 \mathrm{ResDev}_{it} + \beta_3 \mathrm{Temp}_{it} + \beta_4 \mathrm{Prec}_{it} + \beta_5 \mathrm{Press}_{it} + \beta_6 \mathrm{Hum}_{it} \\
&\quad + \sum_{h=1}^{3} \left[\gamma_h^{(s)} \sin\left(\frac{2\pi h m_t}{12}\right) + \gamma_h^{(c)} \cos\left(\frac{2\pi h m_t}{12}\right) \right] + \delta_{\mathrm{year}(t)} + \rho_1 y_{i,t-1} + \rho_{12} y_{i,t-12} + \theta A_{it} + \varepsilon_{it}.
\end{aligned} \tag{1}
$$

In equation (1), α is the overall intercept and $b_i \sim \mathcal{N}(0, \sigma_b^2)$ is the municipality-specific random intercept capturing time-invariant unobserved heterogeneity. The structural demographic component is constructed from

$$R_{it} = \log(1 + \mathrm{residenti}_{it}),$$

where $\mathrm{residenti}_{it}$ denotes the resident population in municipality i at month t. We then define

$$\mathrm{ResMean}_i = \frac{1}{T_i} \sum_t R_{it}, \qquad \mathrm{ResDev}_{it} = R_{it} - \mathrm{ResMean}_i,$$

where T_i is the number of monthly observations available for municipality i. Hence, $\mathrm{ResMean}_i$ is the municipality-specific average log population over the sample period, while ResDev_{it} is the within-municipality deviation from that average. The weather variables are ordered in the same way as they appear in equation (1): Temp_{it} is monthly temperature, Prec_{it} is logarithmic monthly precipitation, Press_{it} is monthly atmospheric pressure, and Hum_{it} is monthly relative humidity. The index $m_t \in \{1, \ldots, 12\}$ denotes the calendar month associated with observation t, so the harmonic terms for $h = 1, 2, 3$ represent the first three Fourier seasonal components. The term $\delta_{\mathrm{year}(t)}$ is a year fixed effect, with year(t) denoting the calendar year of month t. The coefficients ρ_1 and ρ_{12} multiply the one-month and twelve-month lags of log electricity consumption, namely $y_{i,t-1}$ and $y_{i,t-12}$. Finally, A_{it} is logarithmic tourist arrivals in municipality i at month t, θ is the corresponding tourism coefficient, and $\varepsilon_{it} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is the idiosyncratic error term.

In the empirical implementation, all numeric regressors are standardized before estimation, so θ can be interpreted as the semi-elastic effect of a one-standard-deviation increase in tourist arrivals on log electricity demand, conditional on municipality heterogeneity, meteorology, seasonality and lagged demand.

The sequential models $M0, M0$-pop$, M1, M2, M3$, and $M4$ are nested restrictions of the same equation, obtained by progressively setting subsets of coefficients equal to zero. This nesting structure makes the tourism contribution identifiable as the incremental improvement that arises when θA_{it} is added on top of the structural, weather, calendar and dynamic blocks.

Model comparisons use likelihood-based fit criteria, mixed-model R^2 measures and numerical diagnostics; robustness is also checked using residualized arrivals. Full validation definitions and results are reported in Appendix **??**.

3.2 Historical tail risk

The tail-risk analysis starts from two shock variables. The first is the electricity residual from the final mixed model,

$$e_{it} = y_{it} - \widehat{y}_{it}^{M4}, \tag{2}$$

where y_{it} is observed log electricity consumption and $\widehat{y}_{it}^{M4}$ is the fitted value implied by equation (1). The second is the tourism residual $A_{it}^{\perp}$ already defined in equation (2). Thus, the pair $(e_{it}, A_{it}^{\perp})$ collects the remaining electricity and tourism shocks after removing the conditional mean dynamics captured by the mean-effect specification.

Since the copula is estimated on marginally standardized shocks, we transform both series into pseudo-observations through empirical ranks:

$$U_{it}^{E} = \frac{\operatorname{rank}(e_{it})}{n+1}, \qquad U_{it}^{T} = \frac{\operatorname{rank}(A_{it}^{\perp})}{n+1}, \tag{3}$$

where n is the number of observations in the sample used for copula estimation. Thus, each margin is empirical-uniform by construction; diagnostics confirming this property are reported in Appendix **??**.

For each candidate family $f \in \mathcal{F}$, with

$$\mathcal{F} = \{\text{Gaussian}, \text{Student-}t, \text{Clayton}, \text{Gumbel}, \text{Frank}\},$$

we estimate the copula parameter vector by maximum likelihood:

$$\widehat{\theta}_f = \arg\max_{\theta \in \Theta_f} \sum_{i,t} \log c_f\left(U_{it}^{E}, U_{it}^{T}; \theta\right), \tag{4}$$

where Θ_f is the admissible parameter space of family f and $c_f(\cdot)$ denotes the corresponding copula density. The selected copula is the family that minimizes AIC, using BIC as a secondary ranking criterion. From the fitted copula we recover the lower- and upper-tail dependence coefficients:

$$\lambda_L = \lim_{q \downarrow 0} \Pr(U_{it}^{E} \le q \mid U_{it}^{T} \le q), \qquad \lambda_U = \lim_{q \uparrow 1} \Pr(U_{it}^{E} > q \mid U_{it}^{T} > q). \tag{5}$$

Here λ_L measures lower-tail co-movement and λ_U measures upper-tail co-movement between electricity and tourism shocks.

To localize risk, we complement the pooled copula with municipality-level co-exceedance measures. For a threshold $q \in \{0.90, 0.95\}$, define the exceedance indicators

$$I_{it}^{E}(q) = \mathbf{1}\{U_{it}^{E} \ge q\}, \qquad I_{it}^{T}(q) = \mathbf{1}\{U_{it}^{T} \ge q\}, \qquad J_{it}(q) = I_{it}^{E}(q) I_{it}^{T}(q). \tag{6}$$

In equation (7), $I_{it}^{E}(q)$ indicates whether the electricity residual falls in the upper tail at threshold q, $I_{it}^{T}(q)$ indicates the corresponding tourism-tail event, and $J_{it}(q)$ indicates joint exceedance.

For each municipality i, we compute the conditional co-exceedance probability,

$$p_i(q) = \Pr\left(U_{it}^{E} \ge q \mid U_{it}^{T} \ge q\right) \approx \frac{\sum_t J_{it}(q)}{\sum_t I_{it}^{T}(q)}, \tag{7}$$

the lift statistic,

$$\text{Lift}_i(q) = \frac{\Pr(U_{it}^{E} \ge q, U_{it}^{T} \ge q)}{\Pr(U_{it}^{T} \ge q)(1-q)}, \tag{8}$$

and the average joint severity,

$$S_i(q) = \frac{1}{\sum_t J_{it}(q)} \sum_t J_{it}(q) \frac{U_{it}^{E} + U_{it}^{T}}{2}, \tag{9}$$

whenever the denominator is non-zero. Before defining the municipality ranking, it is important to clarify the role of weights in this setting. The composite-indicator literature stresses that weighting schemes are rarely identified from the data alone and typically reflect either normative priorities, expert judgement, or calibration to an external target; for this reason, weighting and robustness analysis are central methodological issues rather than innocuous technical details Nardo et al. [2008], Greco et al. [2018]. In our application, no external municipal benchmark such as outages, curtailment events, or network breaches is available to estimate optimal weights directly. We therefore treat the ranking score as a heuristic policy index and choose weights that place the greatest emphasis on the most severe conditional tail events at the 95th percentile, while still retaining information on broader upper-tail co-exceedance at the 90th percentile and on the average severity of joint extremes. This choice is also consistent with the tail-risk literature, where conservative tail-focused indices are often preferred when standard summary measures risk understating extreme co-movements Furman et al. [2015], Joe [2014].

The municipality ranking is then based on the composite score

$$R_i = 0.65\, p_i(0.95) + 0.25\, p_i(0.90) + 0.10\, S_i(0.95). \tag{10}$$

In equation (11), R_i is the municipality-level composite risk score, $p_i(0.95)$ is the conditional probability of an electricity upper-tail event given a tourism upper-tail event at the 95th percentile, $p_i(0.90)$ is the analogous quantity at the 90th percentile, and $S_i(0.95)$ is the average severity of joint exceedances at the 95th percentile. This design separates two objects: the copula captures region-wide tail dependence in the joint residual distribution, while the municipal score identifies where extreme tourism shocks are most likely to coincide with extreme electricity shocks.

The copula family, thresholds, shrinkage estimator, bootstrap intervals and external ranking checks are specified in Appendix **??**. These checks are used to quantify uncertainty and to distinguish a screening index from a calibrated forecast.

3.3 Forecasting

This subsection develops scenario-based forecasts of electricity demand for policy analysis and planning. The forecasting strategy has two blocks. First, we forecast regional tourism demand and allocate it across municipalities using a rule that preserves spatial heterogeneity. Second, we propagate tourism scenarios through the municipality-assigned electricity models to obtain load outcomes and stress indicators, with emphasis on municipalities identified as high-risk in the historical tail analysis.

Tourism forecasting. The tourism forecasting block is organized in three steps: (i) a regional time-series model for total monthly arrivals; (ii) a spatial allocation rule that distributes the regional forecast across the 74 municipalities; and (iii) a residual bootstrap that propagates predictive uncertainty into validation and forward scenarios.

(i) Regional model. Let X_t denote total monthly arrivals in Valle d'Aosta and $x_t = \log(1 + X_t)$. We compared automatic seasonal ARIMA, feed-forward neural networks (NNAR), the mixed-effects models of section 3.1, and a quantile gradient-boosted regression tree (QGBRT) fitted independently at $\tau \in \{0.05, 0.50, 0.95\}$; the four candidates were ranked on the 2025 out-of-sample validation window using MAE, RMSE, MAPE, bias and empirical 90% coverage, and the NNAR specification was selected because it dominates the other benchmarks on every error metric while remaining approximately unbiased (see Appendix C, T able 10). The selected model is an autoregressive feed-forward neural network with p recent lags, P seasonal lags at period $m = 12$, and a single hidden layer with k units [Hyndman and Athanasopoulos, 2021]:

$$\widehat{x}_t = \beta_0 + \sum_{j=1}^{k} \beta_j\, \phi\left(\gamma_{0j} + \sum_{\ell=1}^{p} \gamma_{\ell j}\, x_{t-\ell} + \sum_{s=1}^{P} \delta_{sj}\, x_{t-sm}\right) + \eta_t, \tag{11}$$

where $\phi(\cdot)$ is the logistic activation, (β, γ, δ) are weights estimated by minimizing the in-sample squared error averaged over an ensemble of random initializations, and η_t is the residual. The fitted model is denoted $\text{NNAR}(p, P, k)_m$ with $m = 12$.

(ii) Spatial allocation. The regional forecast is converted into municipality-level arrivals through a share-based rule. This step is needed because policy decisions and network stress are local, while the most reliable forecast is estimated at regional level. A static benchmark (BASE) would keep each municipality at its long-run average share of regional arrivals; this is coherent but too rigid when local composition shifts over time. Our production rule, METHOD F + METEO, is designed to solve that trade-off: first, it builds a robust recent seasonal baseline (to track structural change without overreacting to outliers); second, it applies a weather-based correction (to capture predictable month-specific reallocations across municipalities). The two stages produce, for each comune i and target month t^*, a blended share $\widetilde{s}_{i,t^*}$, which is renormalized to one each month and applied to the regional total, so forecasts remain regionally coherent while allowing realistic local redistribution. The relative weight of the weather correction is governed by a mixing intensity λ, set to $\lambda = 0.10$ (with ε a numerical floor); for validation months, weather inputs are observed, while for the forward horizon they are replaced by monthly meteorological climatology. The full mathematical definition is reported in Appendix B.1, equations (20)–(22).

The municipal forecast is finally obtained by disaggregating the regional projection, i.e. multiplying the blended municipal share $\widetilde{s}_{i,t^*}$ by the regional arrivals forecast $\widehat{X}_{t^*}$ from the NNAR model in (13):

$$\widehat{a}_{i,t^*} \;=\; \widetilde{s}_{i,t^*} \cdot \widehat{X}_{t^*}. \tag{12}$$

(iii) Bootstrap and uncertainty. Predictive intervals are obtained by a block bootstrap on the regional residuals [Bergmeir et al., 2016]. We generate B bootstrap series $\{x_t^{(b)}\}$ via STL decomposition and moving-block resampling, refit (13) on each, and simulate a joint validation–forward path $\widehat{X}_{t^*}^{(b)}$ for t^* spanning the 9-month 2025 validation window and the 12-month forward horizon. Applying (14) to every replicate yields the empirical predictive distribution at both regional and municipal level, from which the 5%, 50% and 95% quantiles and the bagged mean are extracted. The forecasting performance of the BASE and METHOD F + METEO rules is compared on the 2025 validation block using MAE, RMSE, MAPE, the relative total error, the empirical 90% coverage, and the share of municipalities flagged as implausible.

The choice of METHOD F + METEO as the production rule is itself the outcome of an out-of-sample selection, not an a priori assumption. Conceptually, BASE tends to optimize aggregate fit but can generate implausible allocations in smaller and highly seasonal comuni; more flexible municipality-specific models reduce that rigidity but may overfit idiosyncratic noise. METHOD F + METEO is the compromise that performs best on this bias-variance trade-off: it substantially lowers the implausibility rate (about 6.8% vs. 10.8% for BASE) while preserving comparable aggregate accuracy and regional bias. This is why it is used as the production allocation rule. The full model horse-race and metrics are reported in Appendix B.

Tail-risk forecasting. The forecasting strategy for the tail-risk block is organized in two steps. First, we generate scenario-based forecasts of electricity demand by feeding the tourism forecasts into the mean-effect model. Second, we evaluate the resulting distribution of electricity demand against the historical tail-risk evidence, with particular attention to the high-tourism municipalities identified in the previous step. This step is not a formal forecast validation exercise, since the tail-risk analysis is based on historical co-movements rather than out-of-sample predictive performance, but it provides a forward-looking policy tool that can be used for scenario analysis and planning. In practice, after selecting the municipality-level forecasting model, we replace the tourism input with the scenario path (actual versus q95 arrivals) while keeping the other predictors fixed, and we obtain scenario-based forecasts of electricity demand.

From a mathematical perspective, we can write the mean-effect model as:

$$\begin{aligned}
\hat{y}_{it}^{(m)}(a) &= f_m(z_{it}, a)\,, \qquad m \in \{\text{Mixed}, \text{GAMM}, \text{Prophet}\},\\
\hat{y}_{it}^{\text{base}} &= \hat{y}_{it}^{(m_i)}(a_{it})\,,\\
\hat{y}_{it}^{\text{q95}} &= \hat{y}_{it}^{(m_i)}\left(a_{it}^{95}\right),\\
\hat{L}_{it}^{\text{base}} &= \exp\left(\hat{y}_{it}^{\text{base}}\right) - 1,\\
\hat{L}_{it}^{\text{q95}} &= \exp\left(\hat{y}_{it}^{\text{q95}}\right) - 1,\\
\Delta L_{it} &= \hat{L}_{it}^{\text{q95}} - \hat{L}_{it}^{\text{base}},\\
\Delta L_{it}^{(\%)} &= 100\left(\frac{\hat{L}_{it}^{\text{q95}}}{\hat{L}_{it}^{\text{base}}} - 1\right).
\end{aligned} \tag{13}$$

where m_i is the municipality-level assigned model from the validation-based assignment rule; $a_{it} = \log(1 + \text{arrivals}_{it})$ is observed tourism input; $a_{it}^{95} = \log(1 + \text{arrivals}_{it}^{95})$ is the bootstrap q95 tourism scenario input; and z_{it} collects the non-tourism predictors (meteorology, seasonality, lags and dynamic controls). Hence, the scenario block replaces the tourism predictor while keeping all other predictors unchanged, and computes the implied level and percentage load impact. The baseline load path L_{it}^{base} is therefore the conditional mean forecast from the single model assigned to municipality i, evaluated at the observed arrivals path, rather than an average across all candidate models.

3.4 From Historical Tail Risk to Forward Extreme Stress

This section links the forward tourism-shock forecast from equation (15) to the historical tail-risk diagnostics from equations (8) and (11). Let ΔL_{it} denote the forecasted load shock for municipality i and month t under the q95 tourism scenario, and let $p_i(0.95)$ and R_i be the municipality-specific conditional tail probability and composite risk score estimated on the historical sample.

We define the forecast-conditioned tail-impact at municipality-month level as

$$I_{it}^{\text{tail}} = \Delta L_{it}\, p_i(0.95), \tag{14}$$

which scales the deterministic scenario shock by the municipality's historical propensity to co-exceed in the upper tail. To prioritize municipalities where exposure and vulnerability are jointly high, we further define the weighted impact

$$I_{it}^{\text{w}} = I_{it}^{\text{tail}}\, R_i = \Delta L_{it}\, p_i(0.95)\, R_i. \tag{15}$$

Annual municipality-level stress indicators are obtained by temporal aggregation:

$$\mathcal{I}_i^{\text{tail}} = \sum_{t \in \mathcal{T}} I_{it}^{\text{tail}}, \qquad \mathcal{I}_i^{\text{w}} = \sum_{t \in \mathcal{T}} I_{it}^{\text{w}}, \tag{16}$$

where $\mathcal{T}$ is the forecast horizon (in our application, monthly 2025 scenarios).

The regional monthly indicators are

$$\mathcal{I}_{t,\text{reg}}^{\text{tail}} = \sum_i I_{it}^{\text{tail}}, \qquad \mathcal{I}_{t,\text{reg}}^{\text{w}} = \sum_i I_{it}^{\text{w}}, \tag{17}$$

which summarize, respectively, expected tail-linked stress and risk-weighted tail-linked stress at system level.

Finally, municipalities are ranked by $\mathcal{I}_i^{\text{w}}$ to identify where forward extreme pressure is most policy-relevant. This ranking preserves a clear decomposition between forecasted exposure (ΔL_{it}), historical conditional dependence ($p_i(0.95)$), and structural vulnerability (R_i).

4 Main Results

4.1 Historical Association between Tourism and Electricity Demand

Table 1 reports the key model comparisons. Adding tourism arrivals to the dynamic specification improves fit relative to the baseline dynamic model, and the heterogeneous tourism specification performs best among the current alternatives.

Table 1: Model comparison for the main mean-effect specifications.

Model	AIC	BIC	LogLik	R^2_M	R^2_C
$M3$ base	-3629.79	-3476.12	1837.89	0.9821	NA
$M4$ arrivals	-3703.17	-3542.81	1875.58	0.9823	0.9824
$M4$ arrivals × temperature	-3702.87	-3535.83	1876.43	0.9822	0.9824
$M4$ heterogeneous arrivals	-3741.79	-3568.07	1896.89	0.9824	0.9825

The estimated coefficient on arrivals in the benchmark tourism model is positive and precisely estimated ($\hat{\beta} = 0.0342$, standard error 0.0037). In the heterogeneous model, the baseline arrivals effect remains positive (0.0195), while the interaction with tourism-intensive municipalities is also positive (0.0323), indicating stronger transmission in high-tourism locations.

Two additional points are important for interpretation. First, information criteria show that adding tourism to the dynamic benchmark materially improves fit (AIC from -3629.79 to -3703.17), and allowing heterogeneous transmission improves fit again (AIC to -3741.79). Thus, tourism contributes explanatory power both as an average effect and as a municipality-specific amplifier.

Second, the heterogeneous specification implies a sizable gradient across space: in tourism-intensive municipalities, the effective arrivals effect is the sum of the baseline and interaction terms ($0.0195 + 0.0323 = 0.0518$), about 2.7 times the baseline component alone. This is consistent with local economic structure: where hospitality and seasonal services are more concentrated, a comparable tourism shock generates a stronger electricity-demand response.

The relatively small changes in R^2 across final specifications should be read in context. Because the model already includes strong persistence and seasonal controls, much baseline variation is absorbed before tourism enters; in this setting, economically meaningful tourism effects may appear as modest incremental gains in aggregate fit metrics while remaining statistically and policy relevant.

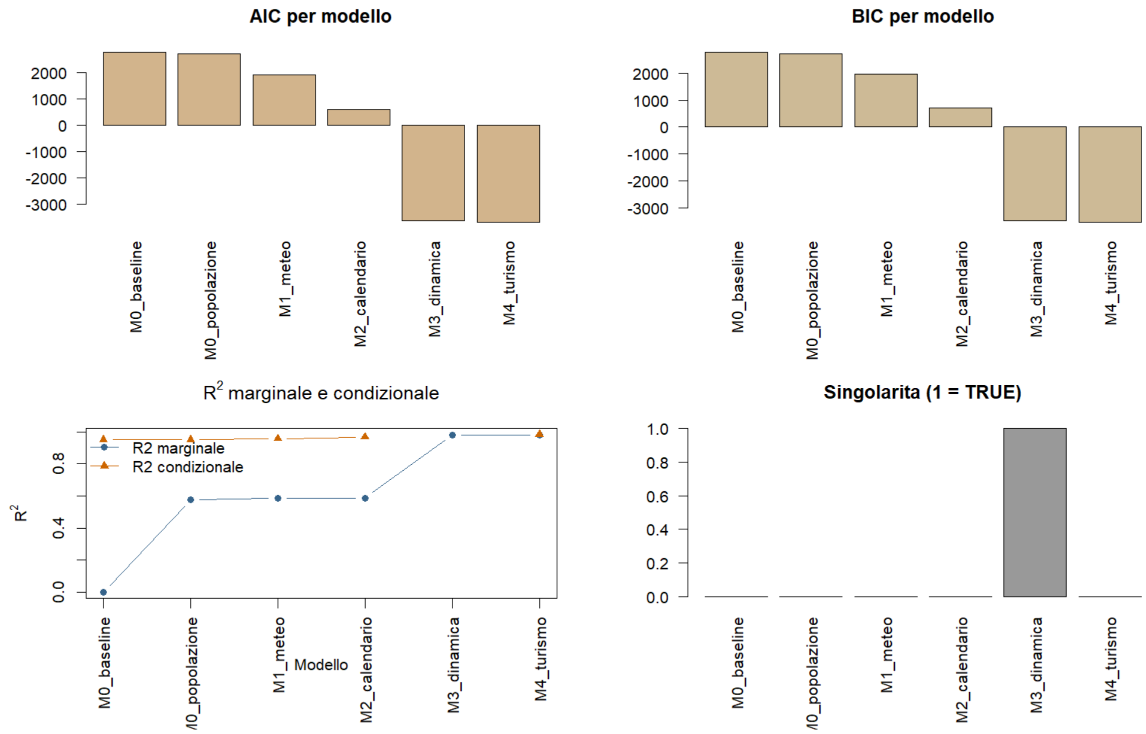

Figure 2: Model comparison across the nested mean-effect specifications.

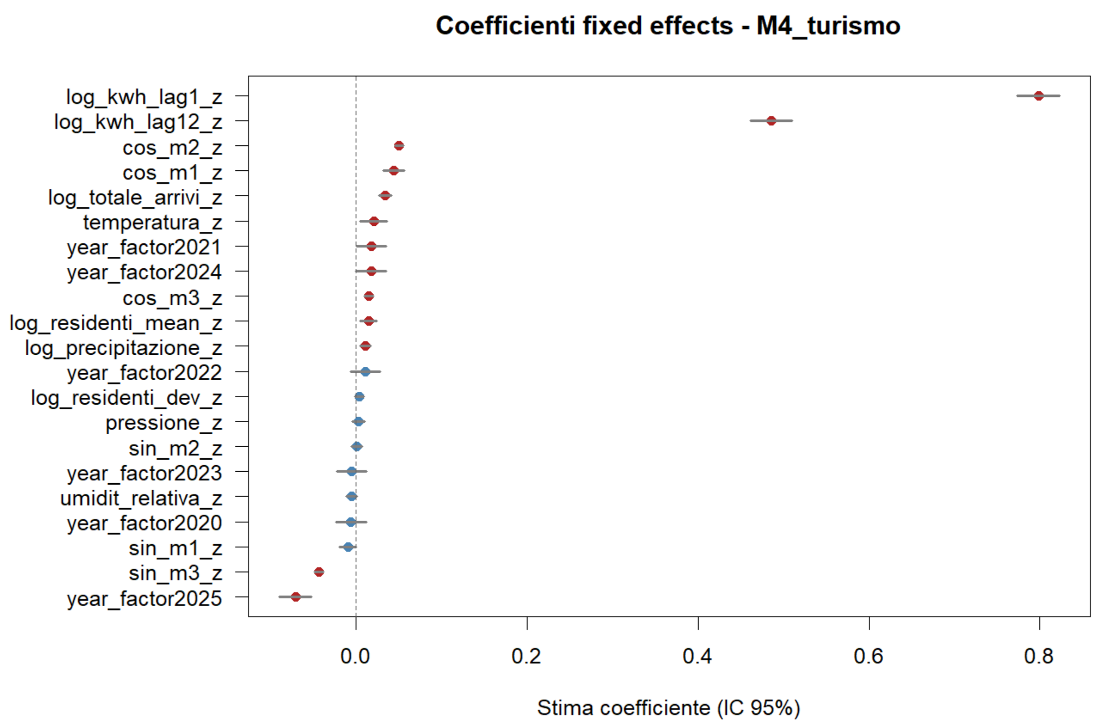

Figure 3: Estimated coefficients for the final tourism specification.

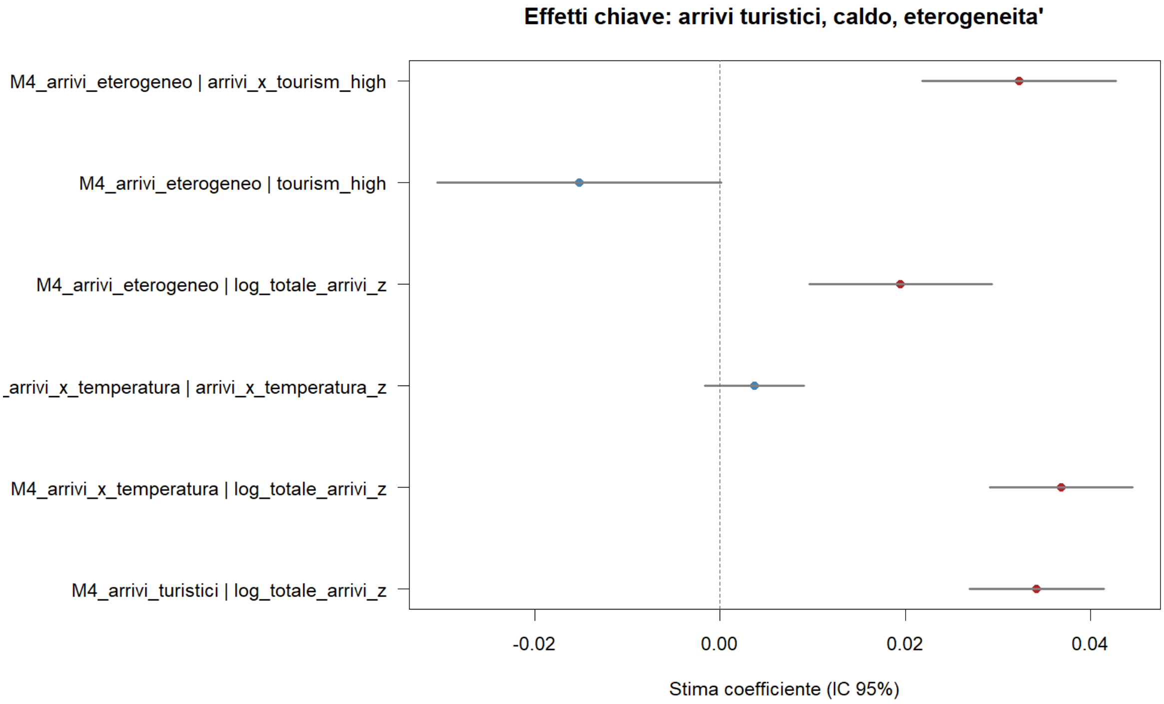

Figure 4: Key coefficients from the tourism extensions: benchmark, interaction with temperature, and heterogeneous tourism effect.

4.2 Tail dependence and localized risk

The pooled residual dependence is best described by a Gumbel copula, indicating asymmetric dependence concentrated in the upper tail. The estimated upper-tail dependence is modest in the pooled sample ($\lambda_U \approx 0.030$), with no evidence of lower-tail dependence. Crucially, a nonparametric pair bootstrap ($B = 1000$) shows that this pooled coefficient is not statistically distinguishable from zero: its 95% percentile interval is $[0.000, 0.014]$, and the pooled Kendall's τ is essentially null and insignificant ($\tau = -0.007$, $p = 0.44$). In plain terms, at the regional level very high tourism shocks are *not* reliably associated with very high electricity shocks once the conditional mean is removed. Once the analysis is localized, the picture changes: upper-tail dependence rises to 0.087 for the top 10 municipalities by empirical tail risk and to 0.065 for the top 10 municipalities by tourism intensity, and in both cases the bootstrap intervals exclude zero ($[0.055, 0.182]$ and $[0.060, 0.178]$, respectively) while Kendall's τ is positive and highly significant ($\tau = 0.089$, $p < 0.001$ in both segments). Hence, the tourism–electricity extreme-event linkage is real but concentrated in specific municipalities rather than being strong and uniform across the whole region.

Table 2: Copula summary by sample segment. The final column reports the 95% percentile bootstrap interval for the upper-tail dependence coefficient ($B = 1000$ pair-resamples). The pooled interval includes zero; both localized intervals exclude it.

Segment	Obs.	Family	LogLik	AIC	BIC	λ_U	λ_U 95% CI
All municipalities	5892	Gumbel	6.491	-10.983	-4.301	0.030	[0.000, 0.014]
Top 10 tail-risk municipalities	804	Gumbel	14.855	-27.711	-23.021	0.087	[0.055, 0.182]
Top 10 tourism-intensity municipalities	804	Gumbel	13.231	-24.461	-19.771	0.065	[0.060, 0.178]

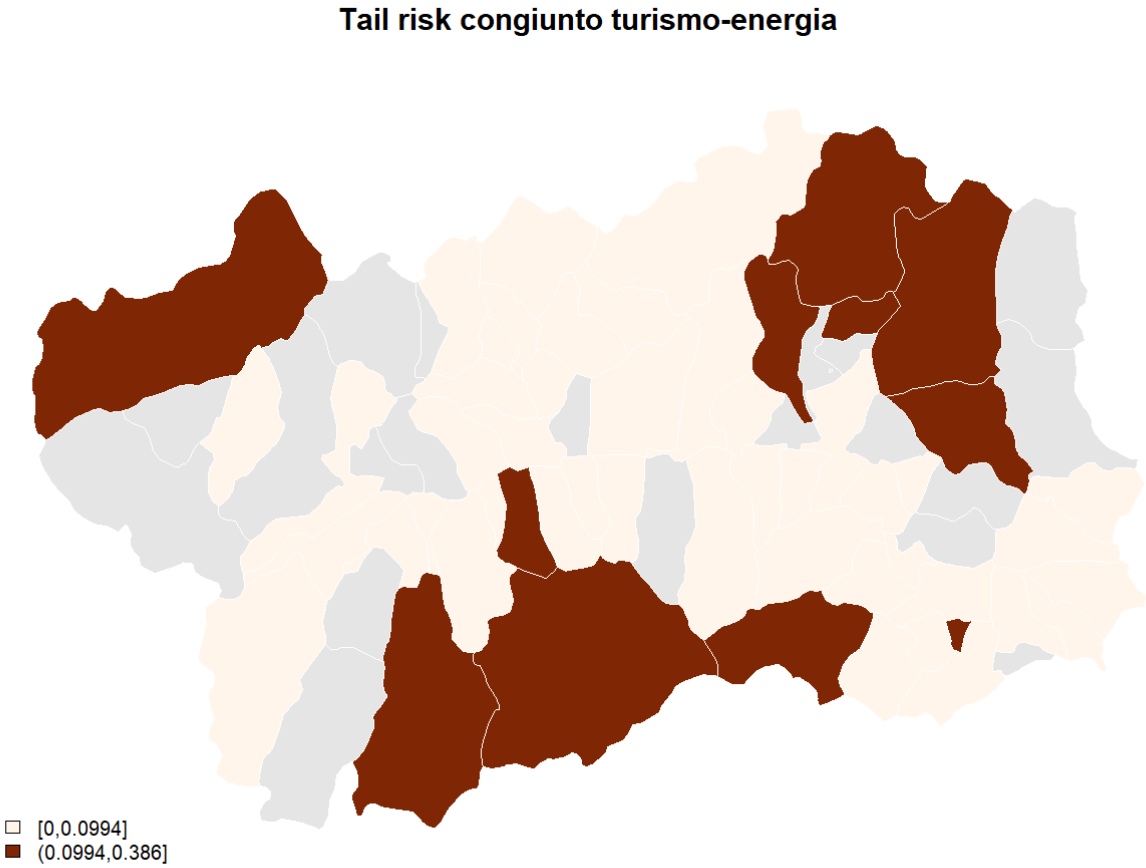

Figure 5: Historical geographical distribution of the municipality-level joint tail-risk score.

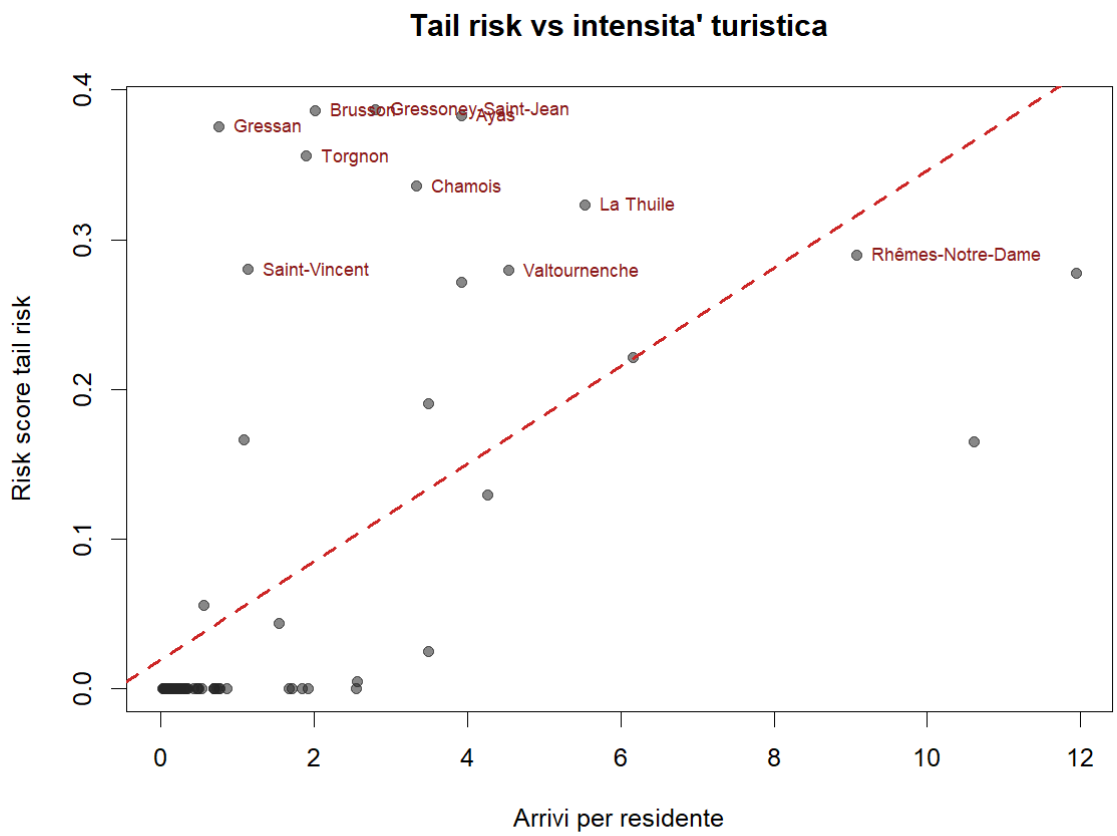

Figure 6: Tail-risk score and tourism intensity. The positive association is clear but not one-to-one.

4.3 Historically most exposed municipalities

As reported in Table 3, the tail-risk ranking identifies a small group of municipalities where extreme tourism and electricity shocks tend to coincide more often than elsewhere. In practical terms, these are the places where a tourism surge is most likely to translate into an electricity-stress episode, because the local system combines high exposure with a non-negligible probability of co-extreme residual shocks. The leading municipalities are Gressoney-Saint-Jean, Brusson, Ayas, Gressan, Torgnon, Chamois, La Thuile and Rhêmes-Notre-Dame. Some of them are highly tourism-intensive, but the overlap is not complete: only four municipalities appear in both the top-10 tail-risk list and the top-10 tourism-intensity list, so tourism intensity alone does not fully explain vulnerability. Figure 6 shows that the correlation between the tail-risk score and tourism intensity is positive but not perfect, which confirms that tourism is an important driver of risk but not the only one. In practise, there many municipality in which the risk score is elevated, but the tourism intensity is not. For example, Gressan has a risk score of 0.376 but only 0.76 arrivals per resident, while Chamois has a risk score of 0.336 but 3.33 arrivals per resident. The correlation between the empirical risk score and arrivals per resident is approximately 0.61, which means that tourism pressure is an important driver of risk, but local tail dependence and municipality-specific structure still matter materially.

Table 3: Top municipalities by empirical tail risk. The R_i 95% CI is the percentile interval from a clustered (municipality-block) bootstrap ($B = 1000$), and P(top-10) is the bootstrap probability that the municipality falls in the regional top-10. Data-poor municipalities (few joint-95 events) have very wide intervals and lower inclusion probabilities.

Municipality	Risk score	$P(E \mid T)_{95}$	R_i 95% CI	P(top-10)	Arr./res.
Gressoney-Saint-Jean	0.387	0.333	[0.019, 0.853]	0.64	2.80
Brusson	0.386	0.333	[0.000, 0.915]	0.62	2.02
Ayas	0.383	0.333	[0.031, 0.788]	0.86	3.93
Gressan	0.376	0.333	[0.000, 0.996]	0.62	0.76
Torgnon	0.356	0.333	[0.000, 0.872]	0.62	1.90
Chamois	0.336	0.250	[0.000, 0.729]	0.64	3.33
La Thuile	0.323	0.222	[0.185, 0.490]	0.92	5.53
Rhêmes-Notre-Dame	0.290	0.154	[0.186, 0.416]	0.87	9.08
Saint-Vincent	0.280	0.200	[0.013, 0.494]	0.76	1.13
Valtournenche	0.280	0.143	[0.057, 0.433]	0.76	4.54

The two right-most columns of Table 3 make the fragility of the raw ranking explicit. Municipalities whose score rests on only one or two joint exceedances at the 95th percentile—Gressoney-Saint-Jean, Brusson, Gressan, Torgnon and Chamois—have extremely wide risk-score intervals (for example $[0.000, 0.915]$ for Brusson) and top-10 inclusion probabilities near 0.62, meaning that their leading position is not robust to resampling. By contrast, La Thuile and Rhêmes-Notre-Dame, which accumulate many more tail observations, have much tighter intervals and inclusion probabilities above 0.87. This motivates the empirical-Bayes correction reported next.

4.4 Shrinkage and ranking stability

Because the raw conditional probabilities $p_i(0.95)$ are estimated from very few tail events, we regularize them with the empirical-Bayes random-intercept logit model in equation (12). The fitted regional baseline is $\widehat{\pi}^{\mathrm{EB}} = 0.116$, and each municipality is pulled toward this value in proportion to how little tail evidence it carries. The effect on the ranking is substantial precisely where it should be: municipalities whose raw score reflected a single joint exceedance out of three tourism-tail events ($p_i(0.95) = 1/3$) are shrunk back toward the regional mean, while municipalities with more exceedance evidence rise. After shrinkage, the top of the ranking is led by Rhêmes-Notre-Dame, La Thuile and Cogne—all data-rich municipalities—whereas Gressan, Torgnon and Saint-Vincent leave the top-10 and Cogne, Gressoney-La-Trinité and Champorcher enter it. Seven of the original top-10 remain, confirming that the broad message is stable while the precise ordering of data-poor municipalities is not. Table 4 summarizes the largest movements.

Table 4: Effect of empirical-Bayes shrinkage on the tail-risk ranking. p_i^{raw} and p_i^{EB} are the raw and shrunk conditional probabilities $p_i(0.95)$; T_i is the number of tourism upper-tail events. Municipalities with few events are pulled toward the regional baseline 0.116 and drop in rank.

Municipality	T_i	p_i^{raw}	p_i^{EB}	Rank (raw)	Rank (EB)
Rhêmes-Notre-Dame	26	0.154	0.125	8	1
La Thuile	18	0.222	0.137	7	2
Cogne	8	0.125	0.116	12	3
Ayas	6	0.333	0.132	3	6
Gressoney-Saint-Jean	3	0.333	0.124	1	7
Gressan	3	0.333	0.124	4	11
Torgnon	3	0.333	0.124	5	13

4.5 Forward extreme stress under tourism q95 scenarios

Using the integration framework in equations (16)–(19), we combine the forecasted tourism shock with historical municipality-level tail-risk dependence to quantify forward stress for 2025. At regional level, total assigned load increases from 252.05 GWh in the baseline path to 257.15 GWh under the q95 tourism scenario, implying an annual shock of 5.10 GWh (+2.03%). The corresponding tail-conditioned impact is 0.857 GWh, and the risk-weighted impact is 0.279 GWh.

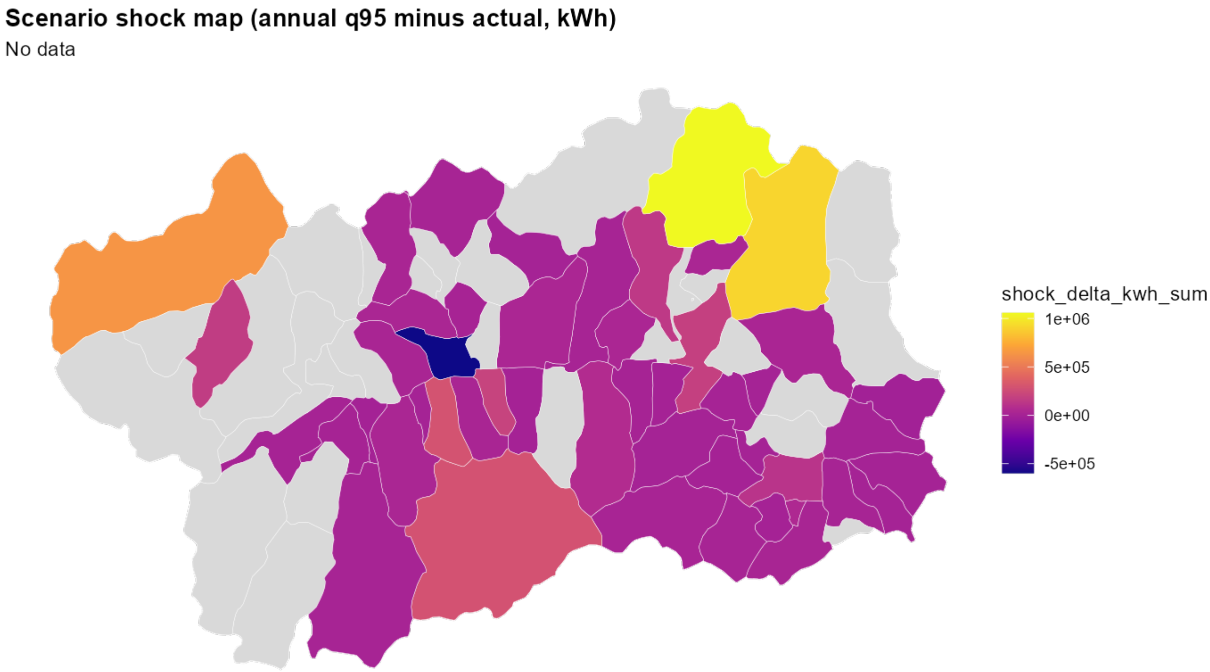

Figure 7: Annual scenario shock under q95 tourism arrivals, measured as q95 minus actual load.

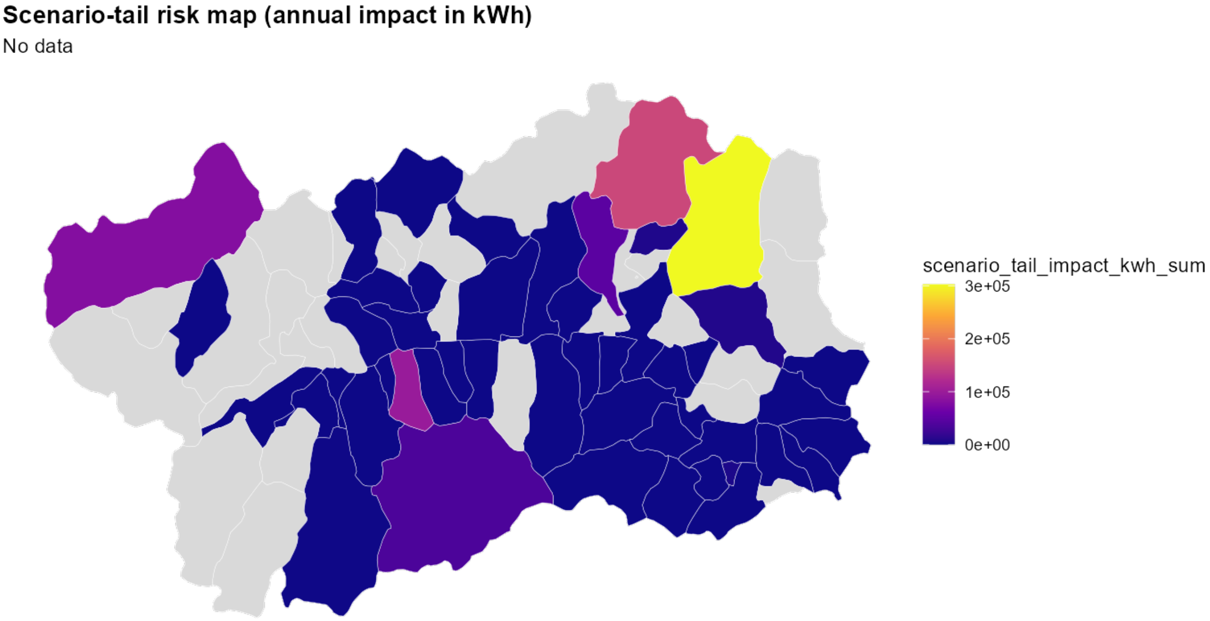

Figure 8: Annual scenario-tail impact after weighting the load shock by historical conditional tail dependence.

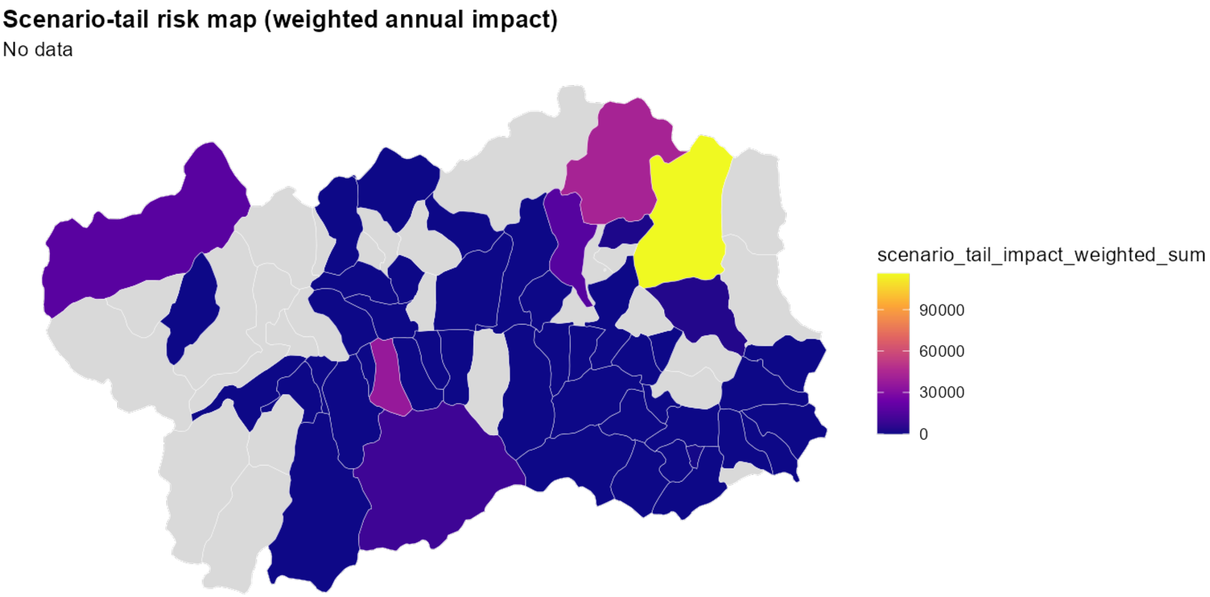

Figure 9: Annual risk-weighted scenario-tail impact combining exposure, tail dependence and municipality vulnerability.

Stress is concentrated in winter and late summer months. The largest monthly regional shock occurs in February 2025 (916.3 MWh, +2.91%), which is also the month with the highest weighted tail-impact (54.4 MWh). September shows the highest percentage shock (+3.62%), indicating that tail-relevant pressure is not purely a winter phenomenon.

Table 5: Regional scenario summary by month for 2025.

Month	Shock (MWh)	Weighted tail impact (MWh)	Shock (%)
Jan	583.3	35.1	1.55
Feb	916.3	54.4	2.91
Mar	673.2	38.4	2.11
Apr	461.1	23.7	1.71
May	282.9	13.6	1.20
Jun	207.3	10.6	0.87
Jul	515.0	26.6	1.95
Aug	653.8	33.1	2.35
Sep	811.3	43.1	3.62

At municipality level, the forward weighted ranking is led by Ayas, Valtournenche, Gressan, and Saint-Vincent, with Courmayeur and Torgnon immediately after. This pattern is consistent with the historical tail-risk evidence but not identical to a pure tourism-intensity ordering, confirming that exposure and dependence jointly shape forward stress.

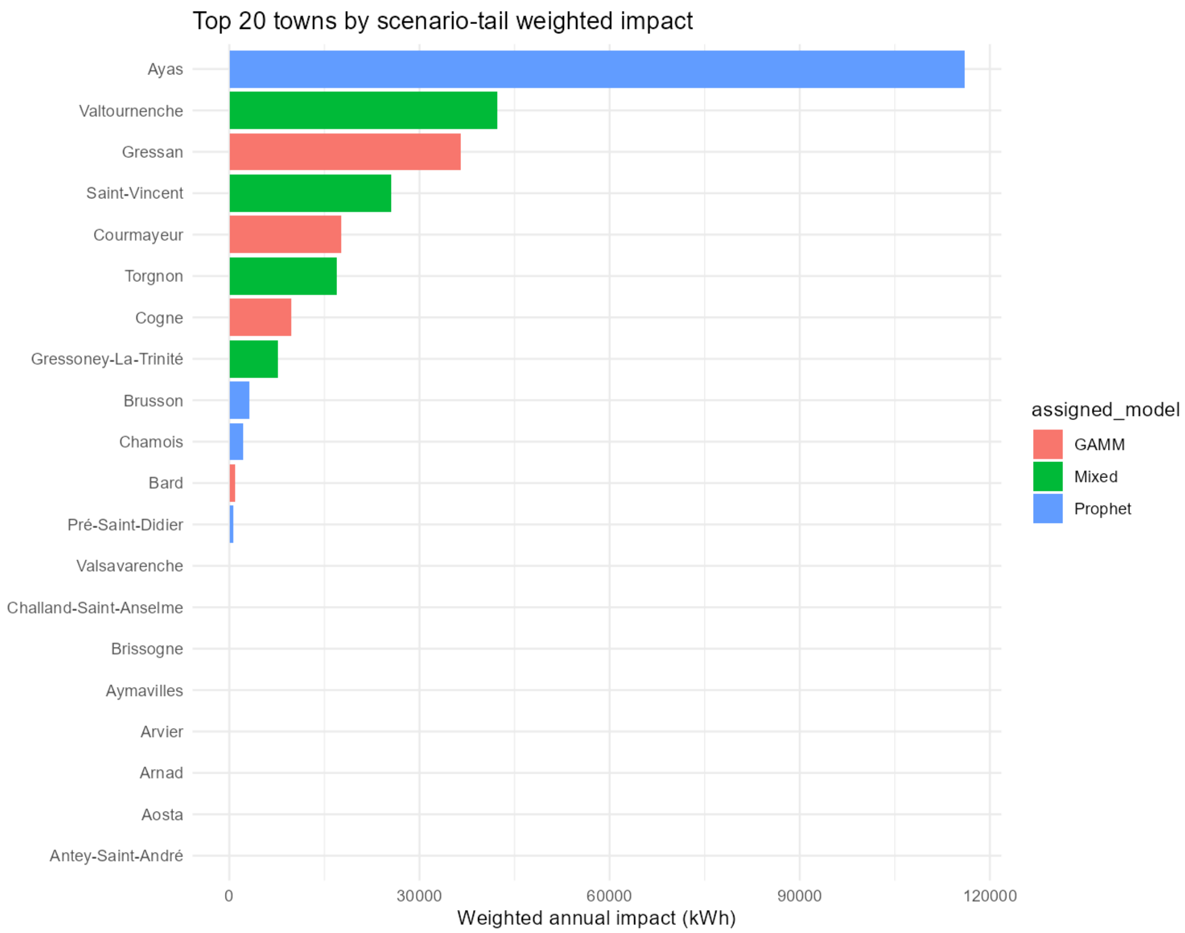

Figure 10: Top 20 municipalities by weighted scenario-tail impact.

Table 6: Top municipalities by weighted scenario-tail impact in 2025. The table reports the municipality risk score with its clustered-bootstrap 95% interval, the bootstrap probability of falling in the regional top-10, the conditional upper-tail probability $P(E \mid T)_{95}$, the annual tail impact in MWh, and the annual weighted impact in MWh.

Municipality	Risk score	R_i 95% CI	P(top-10)	$P(E \mid T)_{95}$	Tail impact (MWh)	Weighted impact (MWh
Ayas	0.383	[0.031, 0.788]	0.86	0.333	302.8	116.
Valtournenche	0.280	[0.057, 0.433]	0.76	0.143	151.0	42.
Gressan	0.376	[0.000, 0.996]	0.62	0.333	97.2	36.
Saint-Vincent	0.280	[0.013, 0.494]	0.76	0.200	91.1	25.
Courmayeur	0.221	[0.019, 0.355]	0.50	0.125	79.9	17.
Torgnon	0.356	[0.000, 0.872]	0.62	0.333	47.5	16.

The columns in Table 6 summarize four distinct layers of the scenario ranking. The risk score is the historical municipality-level composite from the tail-risk block; $P(E \mid T)_{95}$ is the probability that an electricity upper-tail event occurs when a tourism upper-tail event is observed; the tail impact converts the q95 load shock into annual MWh; and the weighted impact further down-weights municipalities with weaker historical tail transmission. As a result, Ayas is the most exposed municipality not only because its raw scenario shock is large, but also because its historical tourism-to-electricity tail linkage is strong.

To assess whether this forward indicator is informative for realized stress, we compare the municipalities ranked highest by predicted weighted scenario-tail impact with the municipalities that actually exhibit the largest positive stress residuals in 2025. Here realized risk is measured at municipality-month level as the positive part of the gap between observed load and the baseline assigned load, namely $r_{it}^{\text{real}} = \max\big(L_{it}^{\text{actual}} - L_{it}^{\text{base}}, 0\big)$, where L_{it}^{base} is the load obtained under the actual-arrivals baseline path. Negative gaps are set to zero because the object of interest is excess stress above the expected baseline, not load shortfalls. Here k is the cut-off size of the ranked list, so $k = 10$ means that the ten highest-ranked municipalities in the predicted list are compared with the ten highest-ranked municipalities in the realized list. Rank association is moderate but clearly positive (Spearman $\rho = 0.315$ for weighted impacts; Pearson $r = 0.285$). In top-k targeting terms, overlap is 6/10 for $k = 10$ (precision/recall 0.60), and average monthly hit rate for top-10 municipalities is 0.467. These diagnostics suggest that the scenario-tail indicator is useful as a prioritization tool, while still reflecting non-negligible uncertainty at municipality-month resolution.

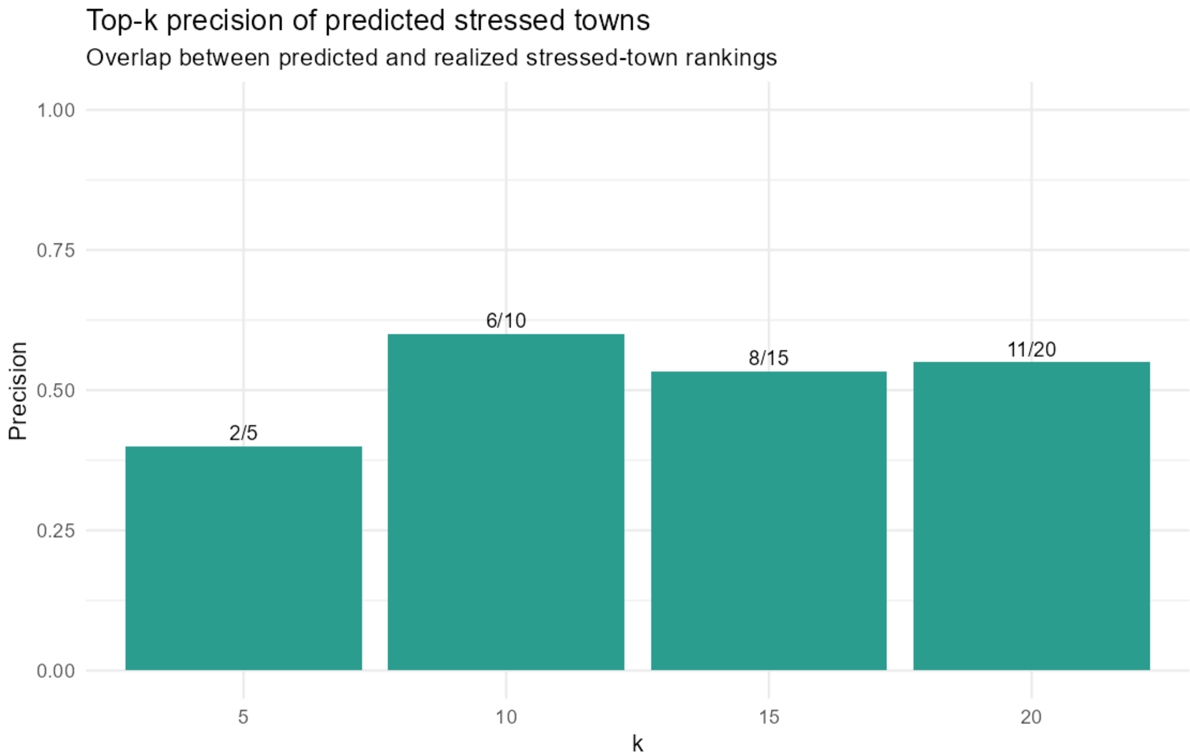

Figure 11: Top-k precision validation for predicted versus realized stressed municipalities. The predicted set is the ranking by weighted scenario-tail impact, while the realized set is the ranking by positive 2025 stress residual. Each curve shows how much of the predicted stressed set is also found among realized stressed municipalities as k grows.

Figure 11 shows that the ranking is informative but not perfect: the top-10 overlap is $6/10$ and the top-20 overlap is $11/20$, so the model recovers a substantial share of the realized stress concentration while still missing some municipality-month episodes. Together with the 0.467 average monthly hit rate, this supports using the scenario-tail ranking as a screening and prioritization device rather than as an exact point forecast of stress.

4.6 Robustness of the composite score and benchmark comparison

Two further exercises probe how much the ranking depends on its construction and whether it carries information beyond simpler indicators. First, we recompute the ranking under alternative weighting schemes. Table 7 shows that the production ranking is highly robust to the choice of weights: equal weights, $q90$-only weights, and even 500 random weight vectors drawn from a symmetric Dirichlet on the simplex all produce rank correlations above 0.98 with the production ordering and top-10 overlaps of at least $8/10$ on average. Only the extreme tail-only and severity-only schemes move the ranking appreciably (rank correlations of 0.90 and 0.89). The composite score is therefore not an artefact of one particular weight vector.

Table 7: Weight-sensitivity of the tail-risk composite score. Each row recomputes R_i under a different weighting of $(p_i(0.95), p_i(0.90), S_i(0.95))$ and reports the Spearman rank correlation and top-10 overlap with the production weights $(0.65, 0.25, 0.10)$. The random-simplex rows summarize 500 Dirichlet draws.

Weighting scheme	Rank corr. vs production	Top-10 overlap
Production $(0.65, 0.25, 0.10)$	1.000	10
Equal $(1/3, 1/3, 1/3)$	0.999	9
Tail-only $(1, 0, 0)$	0.902	9
$q90$-only $(0, 1, 0)$	0.982	8
Severity-only $(0, 0, 1)$	0.892	7
Random simplex (mean)	0.996	8.7
Random simplex (5th–95th pct.)	0.986–1.000	8–10

Second, we benchmark the composite score against four competing rankings, scoring each against realized 2025 positive electricity stress (Table 8). The result is deliberately reported in full because it is informative about the score's intended use. When the target is the *absolute* magnitude of realized stress, a simple historical-average-demand ranking performs best (Spearman 0.80), because absolute stress is dominated by municipality size. The proposed R_i, which is by construction a size-normalized tail-dependence index rather than a demand-level predictor, attains a lower Spearman correlation (0.33) against absolute stress. This is the honest and expected outcome: R_i is designed to flag where tourism and electricity extremes *co-move*, not to forecast which municipality consumes the most energy. Its added value is the localized tail-dependence information in Table 2, which no demand-level benchmark captures.

Table 8: Benchmark comparison of five municipality rankings against realized 2025 positive electricity stress. Higher Spearman and overlap indicate better recovery of the absolute realized-stress ordering. The size-driven historical-average-demand benchmark leads on absolute stress, whereas R_i is a size-normalized tail-dependence index (see text).

Ranking indicator	Spearman vs realized	Top-10 overlap
Historical average demand	0.798	7
Electricity-only extreme frequency	0.505	6
Tourism intensity (arrivals)	0.428	7
Proposed score R_i	0.326	5
Arrivals per resident	0.075	6

Finally, propagating the existing B forecast bootstrap replicates through the tail-impact mapping of equations (16)–(17) yields wide predictive intervals for the risk-weighted regional impact whose 95% bands comfortably include zero. This reinforces the interpretation of the forward ranking as a prioritization screen subject to substantial predictive uncertainty rather than a calibrated point forecast.

5 Discussion

Three substantive messages emerge. First, tourism is robustly associated with the conditional mean of electricity demand in a policy-relevant way. Because the design is observational, this association is interpreted as a conditional-mean relationship after adjustment for the included covariates, not as a causal effect. Second, the residual dependence structure of extremes is not strong at the pooled regional level, which suggests that much of the tourism–energy relationship is already absorbed by the mean-effect specification. Third, once the analysis is localized, extreme co-movements become more visible. This implies that tail risk is not a uniform regional phenomenon but a municipality-specific vulnerability.

These findings also help frame the interpretation of the Gumbel copula. A Gumbel fit implies asymmetric upper-tail dependence: when tourism shocks are unusually high, the probability of a simultaneous unusually high electricity shock is elevated relative to independence, but the magnitude remains limited in the aggregate sample. This is consistent with a system where tourism mainly affects average demand and only locally amplifies extreme demand pressure.

Three caveats deserve emphasis, and the uncertainty analysis makes them explicit. First, the design is observational: tourism arrivals are not randomly assigned, so all estimates are conditional-mean associations after adjustment for the included covariates, not causal effects. Second, the pooled upper-tail dependence is statistically indistinguishable from zero—its 95% bootstrap interval is $[0.000, 0.014]$ and the pooled Kendall's τ is insignificant—so the extreme tourism–electricity linkage should be claimed only at the localized level, where the bootstrap intervals exclude zero. Third, the municipality ranking is a screening device, not a deterministic ordering. The clustered bootstrap shows that several historically top-ranked municipalities owe their position to a single joint exceedance and carry very wide risk-score intervals and top-10 inclusion probabilities near 0.62; empirical-Bayes shrinkage accordingly pulls these data-poor municipalities back toward the regional baseline ($\widehat{\pi}^{\mathrm{EB}} = 0.116$) and promotes data-rich municipalities such as Rhêmes-Notre-Dame and La Thuile, changing three of the ten top positions. The robust message is the broad concentration of vulnerability in a small set of tourism-exposed municipalities, not the exact rank of any individual one.

6 Conclusion

This paper combines mixed-effects modelling and copula-based tail-risk analysis to study how tourism relates to municipal electricity demand in Valle d'Aosta. The evidence indicates a clear positive association between tourism arrivals and the conditional mean of electricity demand and suggests that this association is stronger in more tourism-intensive municipalities. In contrast, residual tail dependence is weak in the pooled sample but increases in a subset of municipalities characterized by higher joint tourism-energy vulnerability. This points to a pragmatic interpretation: tourism matters broadly for average electricity demand, whereas extreme joint stress is a localised issue.

Building on this evidence, the forecasting and scenario block translates the local tail-risk diagnostics into a forward-looking prioritization tool: a regional NNAR tourism forecast is allocated to municipalities and propagated through the assigned electricity models to produce q95 stress rankings, which recover a substantial share of realized 2025 positive stress (top-10 overlap 6/10) while remaining a screening device rather than an exact point forecast. A natural next step is to extend this framework to the small set of highly seasonal municipalities that no candidate model could forecast reliably, and to richer scenario designs beyond the q95 tourism shock.

Data Availability and Reproducibility

All figures and empirical assets used in this manuscript are generated from the project repository. The paper is designed to read the current empirical outputs stored under `Modelli/meanEffect/report_assets`, `Modelli/Tail Risk/report_assets` and `Modelli/Scenarios/report_assets`, with the supporting forecasting tables under `Modelli/Forecast/report_assets`. The mean-effect, tail-risk and forecasting/scenario blocks are all included in the present draft and can be regenerated from the corresponding scripts in the repository.

References

Marina Badal, Miguel Angel Climent, and Juan Antonio Rossello-Nadal. The impact of tourism on energy consumption: A sectoral analysis for the most visited countries in the world. *Economies*, 11(10):263, 2023.

Mohcine Bakhat and Jaume Rossello. Estimation of tourism-induced electricity consumption: The case study of balearics islands, spain. *Energy Economics*, 33(3):437–444, 2011. doi: 10.1016/j.eneco.2010.12.009.

Christoph Bergmeir, Rob J. Hyndman, and José M. Benítez. Bagging exponential smoothing methods using STL decomposition and Box-Cox transformation. *International Journal of Forecasting*, 32(2):303–312, 2016. doi: 10.1016/j.ijforecast.2015.07.002.

Vincenzo Bianco. Analysis of electricity consumption in the tourism sector: A decomposition approach. *Journal of Cleaner Production*, 248:119286, 2020. doi: 10.1016/j.jclepro.2019.119286.

Edward Furman, Jianxi Su, and Ričardas Zitikis. Paths and indices of maximal tail dependence. *ASTIN Bulletin: The Journal of the IAA*, 45(3):661–689, 2015. doi: 10.1017/asb.2015.18.

Andrew Gelman and Jennifer Hill. *Data Analysis Using Regression and Multilevel/Hierarchical Models*. Cambridge University Press, Cambridge, 2007.

Salvatore Greco, Alessio Ishizaka, Menelaos Tasiou, and Gianpiero Torrisi. On the methodological framework of composite indices: A review of the issues of weighting, aggregation, and robustness. *Social Indicators Research*, 2018. doi: 10.1007/s11205-017-1832-9.

Rob J. Hyndman and George Athanasopoulos. *Forecasting: Principles and Practice*. OTexts, Melbourne, Australia, 3rd edition, 2021. Available at: https://otexts.com/fpp3/.

International Energy Agency. Italy: Countries & regions. IEA country profile webpage, 2025. Available at: https://www.iea.org/countries/italy. Accessed 19 May 2026.

Harry Joe. *Dependence Modeling with Copulas*. Chapman and Hall/CRC, Boca Raton, 2014. doi: 10.1201/b17116.

Michela Nardo, Michaela Saisana, Andrea Saltelli, Stefano Tarantola, Anders Hoffman, and Enrico Giovannini. *Handbook on Constructing Composite Indicators: Methodology and User Guide*. OECD Publishing, Paris, 2008. doi: 10.1787/9789264043466-en.

O. B. Soylu, A. N. Menegaki, and Y. Bayraktar. Energy usage trends in top tourist destinations: The role of tourism and trade. *Energy Sources, Part B: Economics, Planning, and Policy*, 19(1):1–15, 2024.

Terna. Dati statistici 2024 sull'energia elettrica in italia. Terna statistical publication webpage, 2025. Available at: https://www.terna.it/it/sistema-elettrico/statistiche/pubblicazioni-statistiche. Accessed 19 May 2026.

A Appendix: Method validation and copula diagnostics

A.1 Mean-effect validation

For each adjacent pair of nested models $M_k \subset M_{k+1}$ we compute the likelihood-ratio statistic

$$\Lambda_{k,k+1} = 2\{\ell(\widehat{\vartheta}_{k+1}) - \ell(\widehat{\vartheta}_k)\},$$

and compare AIC, BIC, marginal and conditional mixed-model R^2. Numerical stability is assessed using variance inflation factors and checks for singular fitted models. As a specification-robustness check, observed arrivals are replaced by

$$A_{it}^{\perp} = A_{it} - \widehat{\mathbb{E}}\left[A_{it} \mid \text{structure}_{it}, \text{weather}_{it}, \text{calendar}_t\right]. \tag{18}$$

A.2 Uniformity of copula inputs

The copula margins are formed using scaled empirical ranks, $\text{rank}(x)/(n+1)$, with average ranks for ties. This transformation places observations strictly inside $(0, 1)$ and produces empirical-uniform margins by construction. In the pooled sample ($n = 5{,}892$), both margins have minimum 0.00017, maximum 0.99983, mean 0.50000, variance 0.08332, and no tied values. We also inspect marginal histograms and rank plots. For municipality subsets, ranks are recomputed within the subset before copula estimation; the pooled ranks are not reused.

A.3 Tail-risk validation and uncertainty

Candidate Gaussian, Student-t, Clayton, Gumbel and Frank copulas are compared using log-likelihood, AIC and BIC. Tail dependence is assessed parametrically through (λ_L, λ_U) and non-parametrically through Kendall's τ and empirical co-exceedance frequencies. Results are repeated for the pooled panel, the top 10 municipalities by empirical tail risk and the top 10 by tourism intensity, using both 90th- and 95th-percentile thresholds and a minimum sample size of 50.

Because upper-tail events are sparse for some municipalities, the 95th-percentile conditional probabilities are also estimated with the empirical-Bayes model

$$J_i \mid T_i \sim \text{Binomial}(T_i, \pi_i), \qquad \text{logit}(\pi_i) = \mu + b_i, \qquad b_i \sim \mathcal{N}(0, \sigma_b^2). \tag{19}$$

Bootstrap resampling provides percentile intervals for copula tail dependence and municipality scores, together with top-10 and top-20 inclusion probabilities. Finally, weight sensitivity and benchmark comparisons against simpler municipality rankings are reported to assess whether the composite score adds information beyond exposure or demand size.

B Appendix: Mean-effect figures

C Appendix: Municipal allocation comparison

C.1 Technical definition of the spatial allocation rule

Let $i \in \{1, \ldots, N\}$ index the $N = 74$ municipalities and $\tau \in \{1, \ldots, 12\}$ the calendar month. Denote by $a_{i,t}$ observed arrivals in comune i at month t and by $A_t = \sum_i a_{i,t}$ the regional total, so the realized share is $s_{i,t} = a_{i,t}/A_t$. The baseline rule (BASE) sets each future share equal to the historical mean $\bar{s}_i = T^{-1} \sum_t s_{i,t}$, which assumes a stable spatial distribution. To relax this assumption we use METHOD F + METEO, built in two stages.

Stage A – temporally weighted seasonal share. Within a rolling 3-year window $\mathcal{W}_t$ preceding the forecast origin, we compute an exponentially weighted seasonal share using a weighted median:

$$s_{i,\tau}^{\text{F}} \;=\; \text{wmedian}(\{s_{i,u} : u \in \mathcal{W}_t,\ \text{month}(u) = \tau\},\ w_u)\,, \qquad w_u = \alpha_F^{\,t-u}, \tag{20}$$

with halflife of 12 months, i.e. $\alpha_F = 0.5^{1/12}$. The resulting shares are renormalized so that $\sum_i s_{i,\tau}^{\text{F}} = 1$ for every τ.

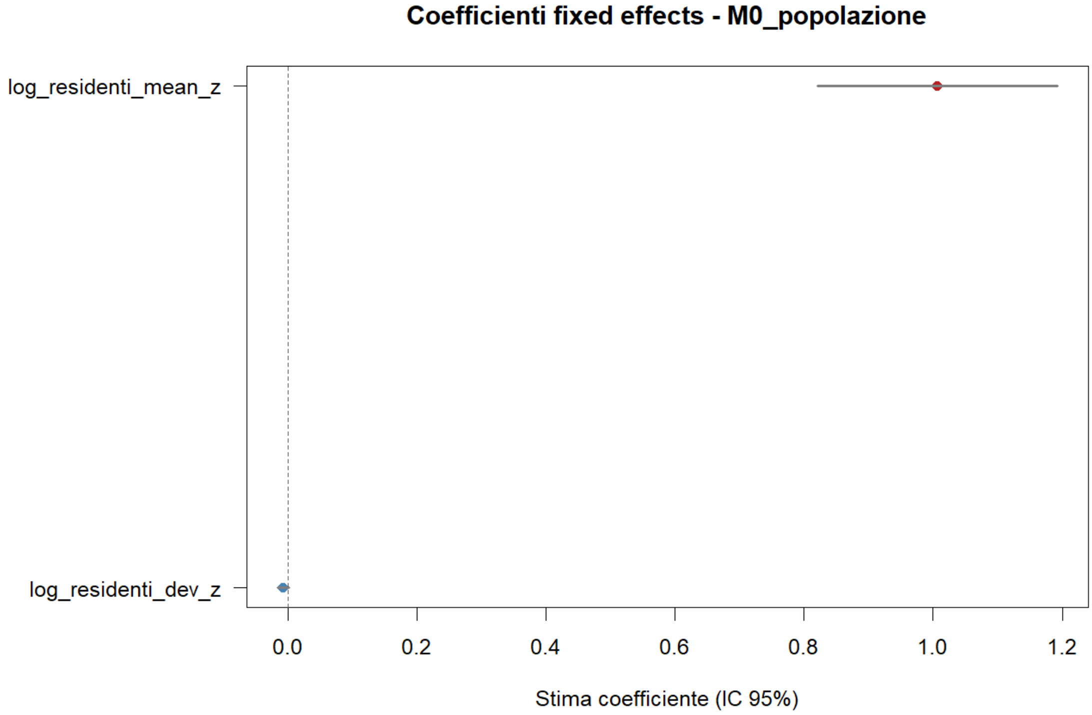

Figure 12: Coefficient plot for the structural baseline model.

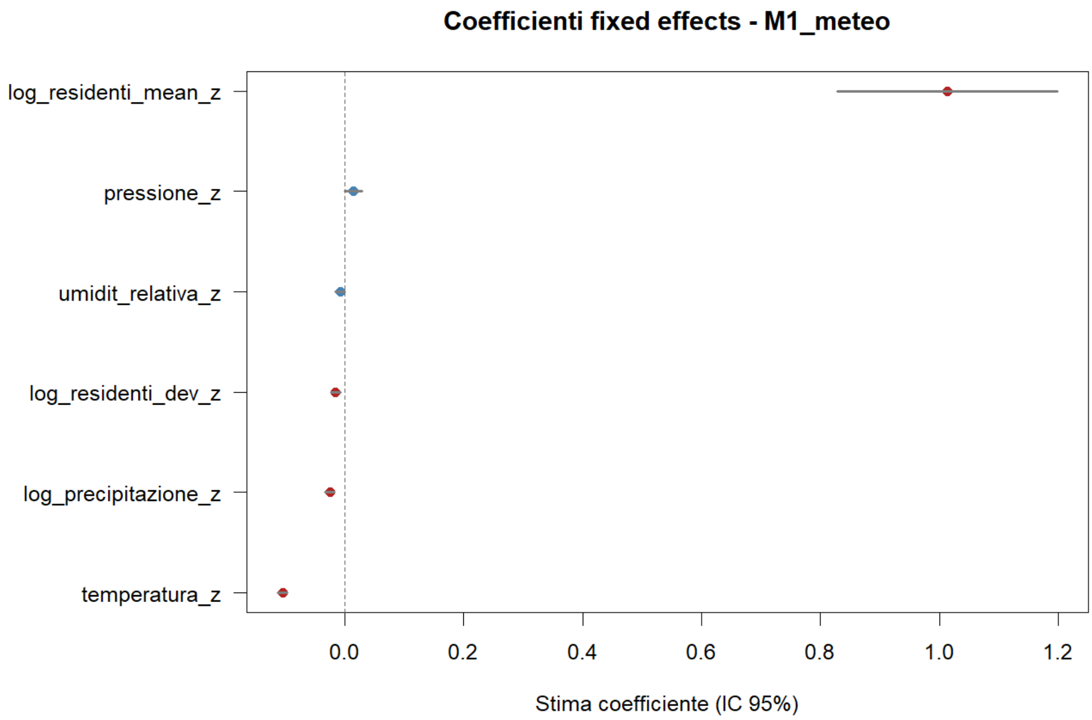

Figure 13: Coefficient plot for the meteorological specification.

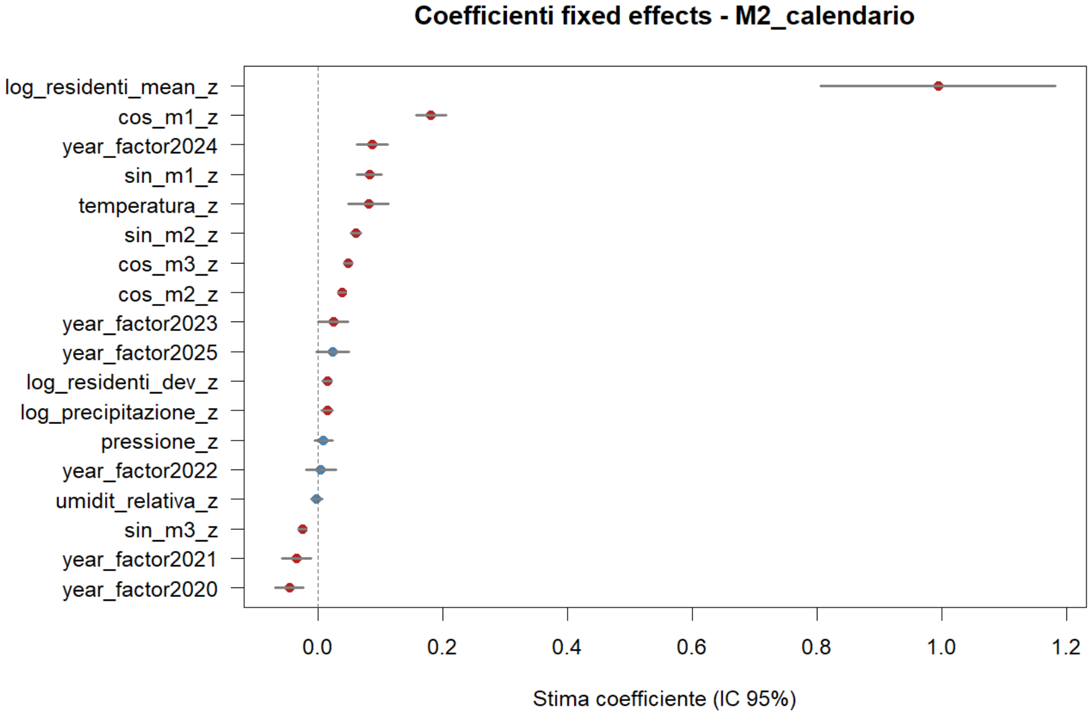

Figure 14: Coefficient plot for the calendar specification.

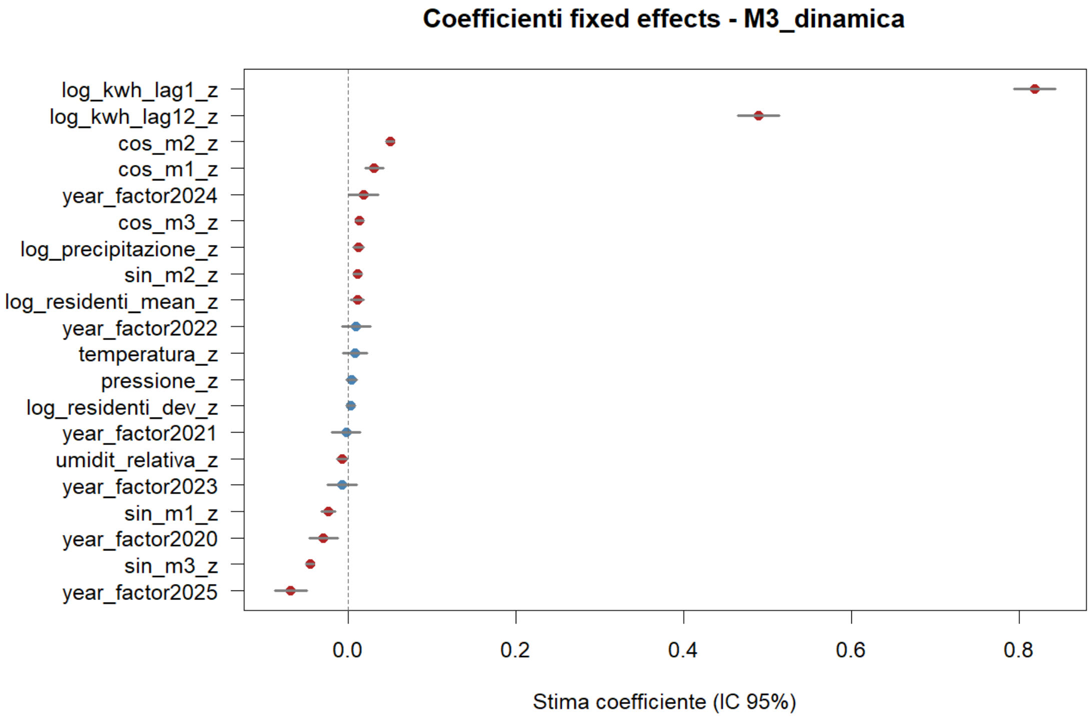

Figure 15: Coefficient plot for the dynamic specification.

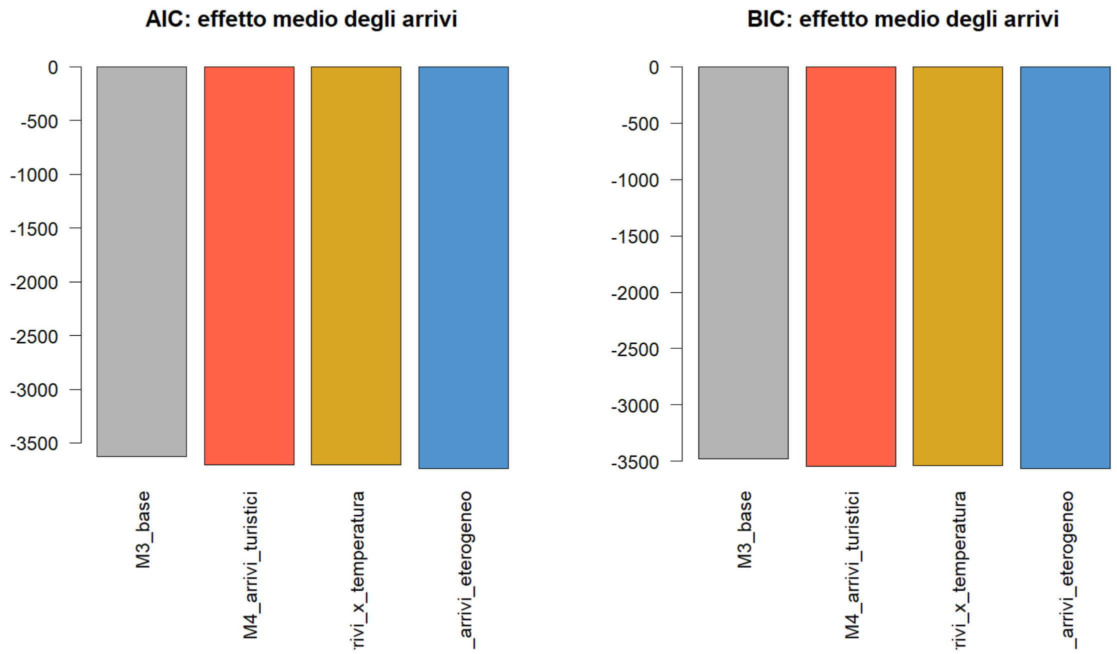

Figure 16: Alternative model comparison for the tourism-focused mean-effect models.

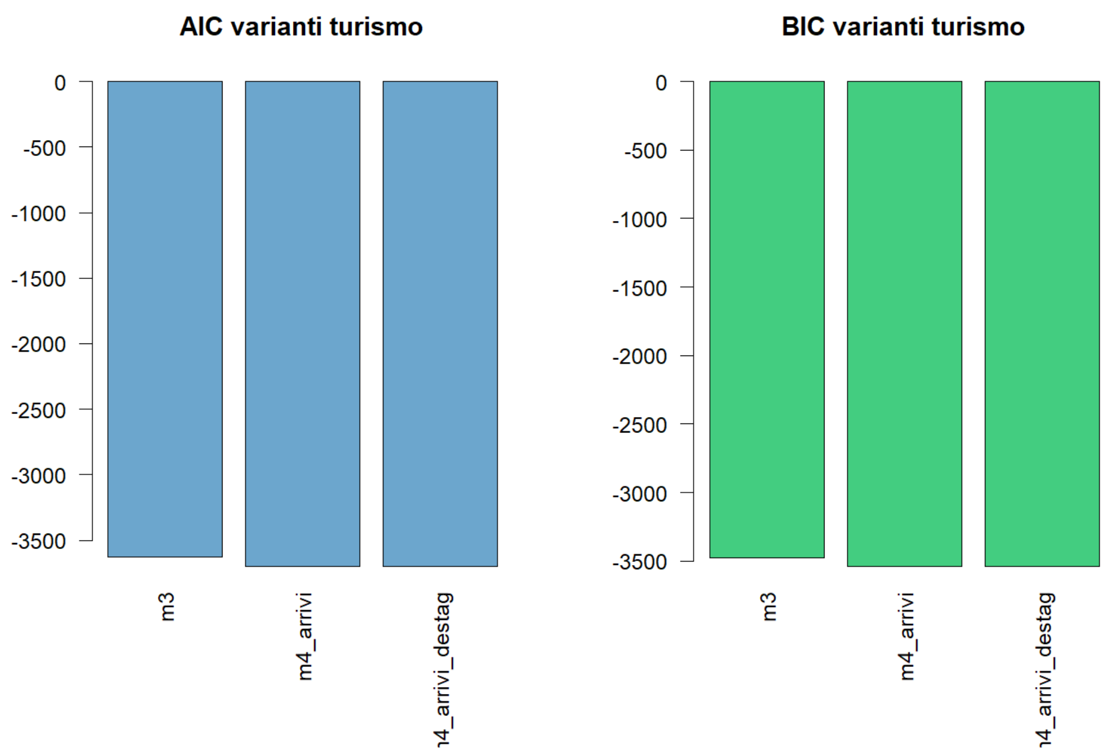

Figure 17: Tourism model comparison diagnostic from the report assets.

Stage B – meteorological adjustment. For each comune i we fit a weather-only weighted least-squares model on the log-share residual, using as covariates $w_{i,t} = (\mathrm{Temp}_{i,t}, \mathrm{Prec}_{i,t}, \mathrm{Press}_{i,t}, \mathrm{Hum}_{i,t})$ and exponentially decayed weights $\omega_t = \alpha_W^{t_0-t}$ with $\alpha_W = 0.7$:

$$\widehat{\boldsymbol{\zeta}}_i \;=\; \arg\min_{\boldsymbol{\zeta}_i} \sum_t \omega_t \left[\log \frac{s_{i,t}}{s^{\mathrm{F}}_{i,\mathrm{month}(t)}} - \boldsymbol{\zeta}_i^\top w_{i,t}\right]^2, \qquad \Delta_{i,t} \;=\; \widehat{\boldsymbol{\zeta}}_i^\top w_{i,t}. \tag{21}$$

The blended share at any target month t^* is then

$$\widetilde{s}_{i,t^*} \;\propto\; \exp\Big\{\log\big(s^{\mathrm{F}}_{i,\mathrm{month}(t^*)} + \varepsilon\big) + \lambda\,\Delta_{i,t^*}\Big\}, \qquad \sum_i \widetilde{s}_{i,t^*} = 1. \tag{22}$$

Table 9 reports the out-of-sample performance of the twelve allocation rules considered for distributing the regional tourism forecast across the 74 municipalities of Valle d'Aosta. All metrics are computed on the nine-month 2025 validation window with the same regional NNAR backbone, so differences across rows are attributable only to the allocation step. The columns are: median and mean municipal MAPE; median and mean of the absolute relative total error $|\hat{A}_i - A_i|/A_i$; total regional MAE; the share of comuni with $|\text{rel. total}| > 0.5$ or > 1.0; the share of comuni with MAPE > 0.75; and the overall implausibility rate (the union of $|\text{rel. total}| > 0.5$ and MAPE > 0.75).

Table 9: Out-of-sample comparison of municipal allocation strategies on the 2025 validation window (74 comuni). Lower values are better in every column. The selected production rule is highlighted in bold. λ refers to the meteo-blend intensity in eq. (22).

Method	Med. MAPE	Med. |rel. tot|	Total MAE	% rel. > 50%	% impl.
BASE static share	0.207	0.097	20 222	4.1	10.8
A: expw + weather	0.322	0.155	38 329	14.9	18.9
B: ETS + reconcile	0.407	0.220	48 528	20.3	25.7
C: BASE + shrunk A (λ=0.10)	0.212	0.094	20 295	5.4	12.2
D: BASE + meteo (λ=0.10)	0.223	0.096	21 183	4.1	10.8
E: NNAR per comune	0.261	0.117	25 185	13.5	23.0
F: expw median 3yr	0.216	0.104	19 991	4.1	8.1
G: adaptive + winsor	0.206	0.096	19 664	2.7	8.1
H: volume shrinkage	0.220	0.097	19 855	5.4	12.2
F + meteo (λ=0.10) [sel.]	0.222	0.097	21 017	4.1	**6.8**
G + meteo (λ=0.10)	0.213	0.093	20 477	2.7	8.1
H + meteo (λ=0.10)	0.230	0.103	20 525	5.4	12.2

The baseline static share is hard to beat on aggregate accuracy because the spatial distribution of arrivals is dominated by a small set of high-volume comuni whose long-run shares are stable. However, BASE produces 10.8% of implausible municipalities, concentrated in small comuni where the static share misses structural changes in the share trajectory. Method A (exp-weighted weather-conditioned shares) and Method B (per-comune ETS + reconciliation) overfit to recent comune-specific volatility and worsen every metric. Per-comune Nnar (Method E) suffers from the same small-sample issue and from the absence of coherence with the regional total. The BASE++ family (F/G/H) improves on BASE by reducing implausibles to 8.1–12.2% while keeping the headline accuracy. Adding the weather-only blend on top of F (F+meteo) is the only configuration that further reduces the implausibility rate to 6.8%, achieving a Pareto improvement over BASE (lower implausibility, comparable median accuracy, comparable regional bias). This is consistent with the diagnostic finding that the residual implausible comuni after F+meteo are five small mountain villages with sparse and very seasonal arrivals, for which no allocation rule based on regional reconciliation is expected to recover the local dynamics. The full per-comune metrics and the bootstrap-based intervals are stored in the project repository under `Modelli/Forecast/report_assets/allocation_comparison_*.csv`.

D Appendix: Regional model comparison

Table 10 reports the out-of-sample performance of the four regional benchmarks evaluated on the same nine-month 2025 validation window (74 comuni aggregated to the regional total). All models are trained on data up to 2024-12-31 and produce 9 monthly forecasts. We compare: (i) Multilevel, the production mixed-effects model summed across comuni; (ii) Auto.arima, an automatic seasonal ARIMA on the regional log-arrivals series; (iii) Nnetar, a feed-forward neural net with one hidden layer and lagged inputs, eq. (13); and (iv) Qgbrt, a quantile gradient-boosted regression tree (`gbm`) fitted independently at $\tau \in \{0.05, 0.50, 0.95\}$ on harmonic and lag features. Metrics are mean absolute error (MAE), root mean square error (RMSE), mean absolute percentage error (MAPE), bias, and empirical 90% interval coverage.

Table 10: Regional model comparison on the 2025 validation window. Lower is better for MAE, RMSE, MAPE and |bias|; coverage 90% should be close to the nominal 0.90. The production regional backbone is highlighted in bold. †Best 1-layer deep NN (BoxCox, 16 units) from independent script.

Model	MAE	RMSE	MAPE	Bias	Cov. 90%
Multilevel (sum of comuni)	63 051	67 977	0.438	−63 051	1.00
Auto.arima	14 609	18 851	0.104	−14 609	1.00
Nnetar (selected)	**9 591**	**11 983**	**0.075**	+1 831	1.00
Qgbrt ($\tau \in \{0.05, 0.50, 0.95\}$)	16 459	23 602	0.096	−14 322	0.56
1-layer deep NN (BoxCox, 16u)†	14 231	21 227	0.096	−10 435	0.44

Nnetar is the most accurate point-forecaster on every error metric (MAPE 7.5%, MAE about 9.6k) and is approximately unbiased, motivating its adoption as the regional backbone for the production pipeline. Auto.arima comes second but exhibits a persistent negative bias, consistent with its inability to fully capture the post-pandemic recovery in arrivals. The naive sum of the per-comune mixed-effects forecasts underperforms strongly at the regional level (MAPE 43.8%): the model is fit to maximise per-comune fit and the comune-level negative biases accumulate when aggregated. The quantile boosted trees achieve a respectable point-forecast accuracy (MAPE 9.6%) but their 90% interval, constructed as $[\hat{q}_{0.05}, \hat{q}_{0.95}]$ from independent quantile fits, covers only 56% of the realised values, reflecting the well-known under-coverage of marginally fitted boosted quantiles on short validation samples. The best 1-layer deep NN (BoxCox, 16 units) from the independent script is included for direct comparison: it is competitive with QGBRT and auto.arima, but does not outperform the production NNAR. The empirical coverage of Nnetar, Auto.arima and the multilevel benchmark is exactly 1.00, indicating that on this nine-month window the parametric bootstrap intervals are conservative; this is expected given the small validation horizon and is in line with results reported by Hyndman and Athanasopoulos [2021] and Bergmeir et al. [2016] on bootstrap-based prediction intervals at monthly frequency. The numbers in Table 10 are reproduced from `Modelli/Forecast/report_assets/tourism_arrivals_model_comparison_metrics.csv` and can be regenerated by running the forecasting script with the lightweight flag `-mode=select`, which skips the bootstrap simulations.

E Appendix: Tail-risk figures

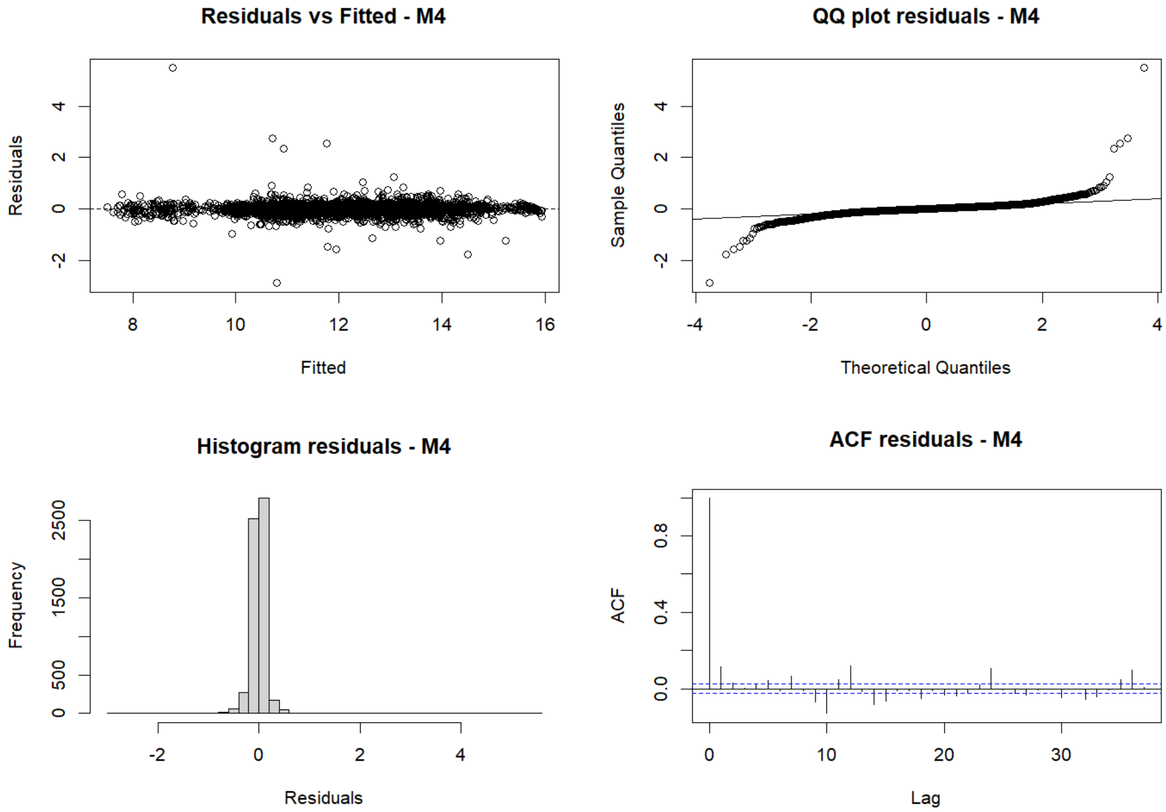

Figure 18: Residual diagnostics for the final mean-effect model.

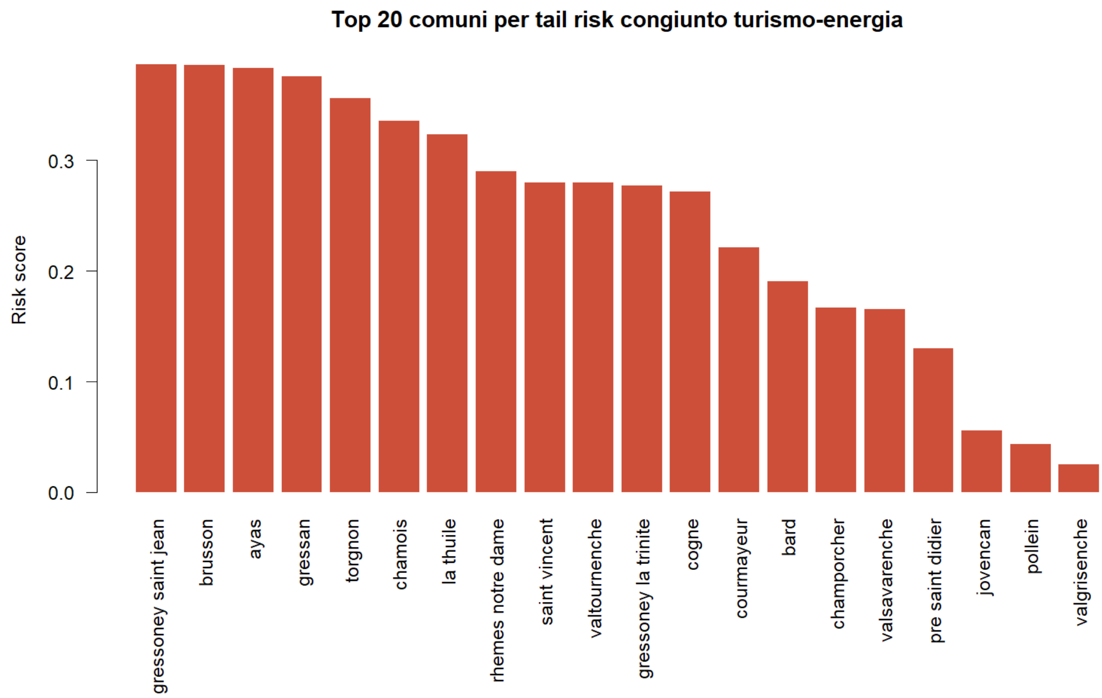

Figure 19: Top 20 municipalities by empirical joint tail-risk score.

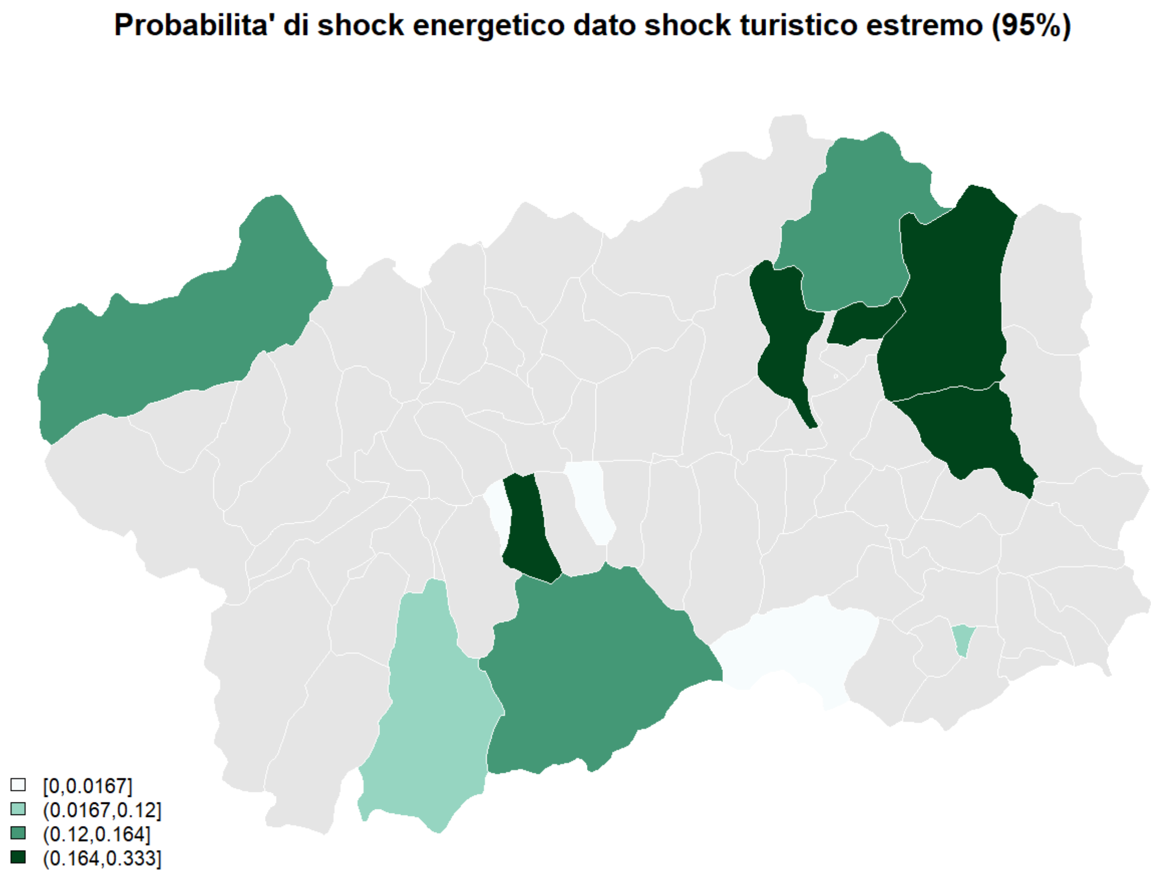

Figure 20: Probability of an extreme electricity residual conditional on an extreme tourism residual at the 95th percentile threshold.

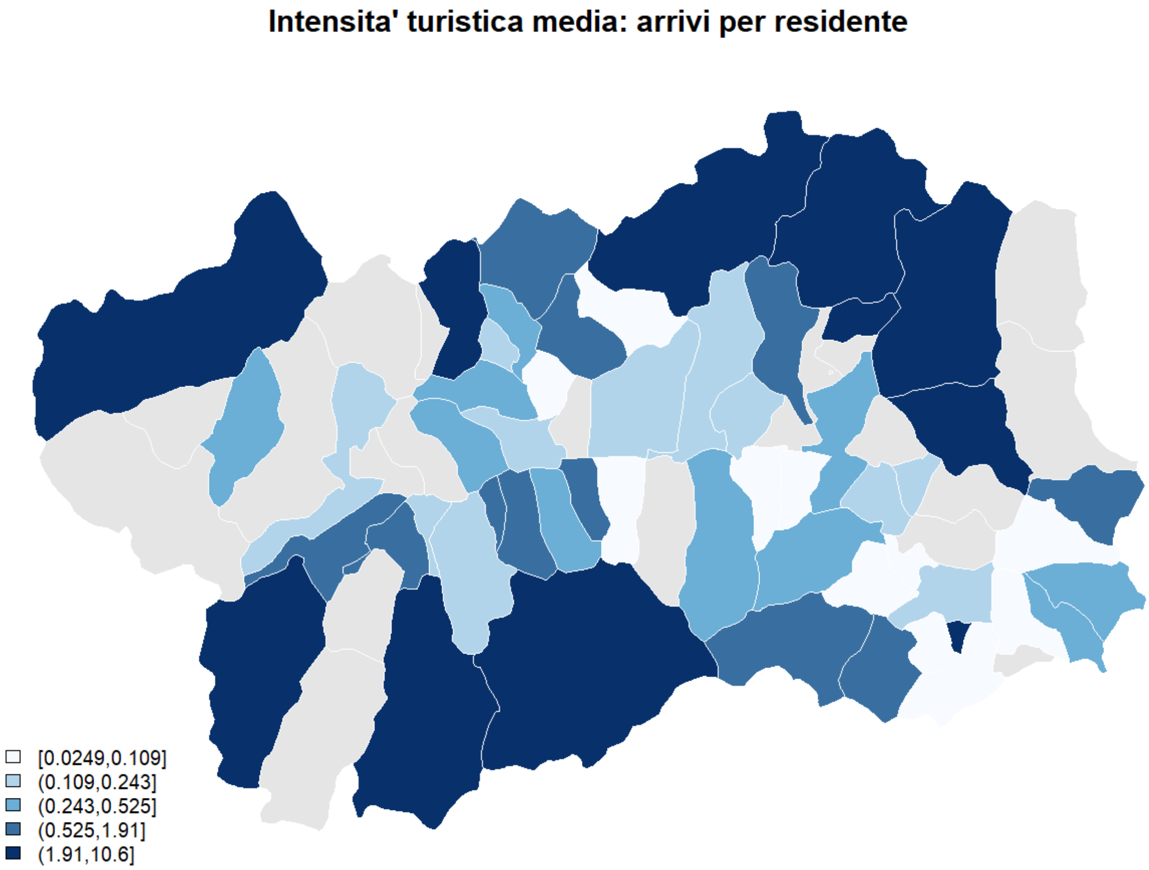

Figure 21: Tourism intensity map measured by arrivals per resident.

F Appendix: Notes for completion

- Add journal-specific front matter, affiliations, acknowledgements and funding statements once the target outlet is fixed.
- If the target journal requires a supplementary appendix, move the full figure archive and extended municipality tables there.